ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN (QSP)
USQ

Sonar Op Submarines (00324.01)

Sonar Op Sub

NQUAL – AILO

MITE IDs
Module 1 – 121993
Module 2 – 108503 (AGSH)
Module 3 – 121994 (OJPR) (AILO)

RECORD OF AMENDMENTS

The training authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC (A)) is the designated development centre responsible for the maintenance of training. Amendments are to be requested through NTDC (A) and approved by CNPTG.

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FOREWORD

1. This training is based on the specification found at :

   a. A-P9-050 Series, Canadian Forces Individual Training and Education System Manuals;

   b. A-PD-055-002/PP-002 Occupation Specification (Sonar Op) dated 1 Sept 12; and

   c. A-PD-055-003/PQ Occupation Specialty Specification (AILO) dated 28 Mar 11.

2. A trainee will achieve the qualification AILO when the standards listed in all Performance Objectives have been met. The qualification AGSH will be granted when the standard listed in Performance Objective 006 have been met.

3. A trainee will achieve the following MITE Course IDs when standards listed in the associated Performance Objectives have been met:

   a. Module 1: POs 001, 002, 003, 004, 005 - 109272
   b. Module 2: PO 006 - 108503
   c. Module 3: POs 001, 002, 003, 004, 005 - 109273

4. The designated TEs for this training is indicated below:

TE	Phase / Module	MITE course ID	NQual
NFS(A) / ADAC	Module 1	109272	AILO
NFS(A)	Module 2	108503	AGSH
Fleet	Module 3	109273	AILO

CHAPTER 1

TRAINING MANAGEMENT DETAILS

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AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Submarine Sonar Operator AILO. This QSP is designed to assist NTDC (A) in developing, Naval Fleet Schools in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with (IAW) NPTORD 4500-22, para 4.5. For this course, a period of instruction is 50 minutes in duration.

Course Duration Calculation	
Allocation of ATT	**Periods**
Sum of all EOs/POs	246
Enabling Checks (ECs)	45
Performance Checks (PCs)	36
Actual Training Time (ATT) Sub-total	41 **trg days**
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22, Table 5.5	**Periods**
Base Administrative Time (BAT)	10
Training Administrative Time (TAT)	10
Training Travel Time (TTT)	0
Physical Fitness (PT)	24
Sub-total	5.5 **trg days**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **400 minutes** **Partial days must be rounded up**	47 **trg days**

REQUIRED COURSE STAFFING

3. Identify key personnel required to conduct the training.

Position	Rank	Qualifications	Qty	PO / EO
Course Officer	Lt (N)	BMOQ, MARS II	1	PO 001 - 006
Senior Instructor	PO2/PO1	BSQ, Sonar Operator RQ PO2, IS, AIT	1	PO 001 - 006
Instructor	MS/PO2/PO1	BSQ, SRWL, AIT	1	PO 001, 002, 004, 005
Trainer Operator (VCSTT)	ANY and civ	VCSTT Operator course	1	PO 001, EO 002.01
Instructor (PAL)	PO2/PO1	UC1, AIT	1	PO 003

Instructor	MS/PO2	Level 1 Vibration Analysis (Offered by Maximum Reliability (MRI))	1	PO 006

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40.

   a. Maximum – 6;

   b. Minimum – 2; and

   c. Optimum – 4.

PREREQUISITES

5. Prerequisites. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

   a. Sonar Operator RQ-S3 (ALFT);

   b. Basic Submarine Qualification (BSQ) Module 1 (AILS - 109201); and

   c. Level 2 Security Clearance or above.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

6. PLARs shall be conducted IAW NPTORD 4500-27, DAOD 5031-1 and CFITES Vol. 12.

SPECIAL INSTRUCTIONS

7. Nil.

CHAPTER 2

PERFORMANCE OBJECTIVES & LESSON GUIDANCE

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PO 001 – Operate Main Frame Sonar

1. **Performance**: Operate Main Frame Sonar

2. **Conditions**:

   a. Given:

      (1) Chronoscope;
      (2) Supervision;
      (3) Sonar; and
      (4) References.

   b. Denied: Nil.

   c. Environment:

      (1) At sea or alongside;
      (2) Night or day; and
      (3) Any weather conditions.

3. **Standard**: The Submarine Sonar Operator shall operate main frame sonar to include:

   a. Performing operator checks on equipment (B0005) to include:

      (1) Setting up passive equipment(s) (T0001);
      (2) Performing communications circuit functional checks (comm checks) (A0005); and
      (3) Operating communications equipment (A0010).

   b. Calculating acoustic sensor range information (C0115) to include:

      (1) Setting up Mainframe Sonar oceanographic equipment (T0005);
      (2) Determining oceanographic effects on submarine operations (C0151); and
      (3) Operating Mainframe Sonar oceanographic equipment (T0006).

   c. Operating active equipment (T0004) to include:

      (1) Setting up main sonar in active mode (T0003);
      (2) Enforcing/Adhering to emission control (EMCON) restrictions (B0020 B0025); and
      (3) Operating mainframe sonar in active mode.

   d. Operating passive equipment(s) (T0002) to include:

      (1) Tracking acoustic contacts (C0070) to include:
         i. Tracking Broadband (T0017);
         ii. Tracking Narrowband (T0018) to include:

(a) Identifying acoustic signatures (T0011); and
(b) Analyzing passive acoustic data (C0065).

iii. Tracking Acoustic Intercept and Ranging (AI&R) (T0016).

(2) Classifying passive acoustic data (C0080).

e. Operate mainframe sonar Noise Monitoring sub-system (T0019);

f. Recording Acoustic Intelligence (ACINT) data (T0020) to include:

(1) Operating sonar logger (T0021); and
(2) Operating Digital Data Collection System (DDCS) (T0022).

g. Presenting oceanographic brief (B0036).

4. **Training Limitations**: Nil.

EO 001.01 - Perform Operator Checks on Equipment (B0005)

Introduction to AN/BQQ-10 Sonar System

Lesson	Conditions	Sequence
001.01A	Classroom	Beginning of PO 001
Time (minutes)	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Explain purpose of BQQ-10 Sonar Systems 2) Describe physical components of BQQ-10 3) Identify fault indicators of BQQ-10 4) Describe system checks of BQQ-10 5) Describe set-up of BQQ-10	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will introduce the students to the purpose, major components, modes of operation, controls, displays and interfaces of the AN/BQQ-10 V7 Sonar Suite. 2) Upon completion of the lecture the instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.01A Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) IETM; and 3) NCPM 403.

Notes to Instructor

The lectures have extensive notes to convey the context of the lesson. Ensure the students have a complete grasp of the layout of this system as it will assist in the understanding of the subsequent lectures.

Short breaks in the presentation will assist in keeping the attention of the students.
Reiterate what this system is used for and the importance of the mainframe sonar in dived navigation
and submarine safety.

EO 001.01 - Perform Operator Checks on Equipment (B0005)

Sonar System Management (SONSYM).

Lesson	Conditions	Sequence
001.01B	Classroom	After 001.01A
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Explain SONSYM purpose; 2) Identify system components; 3) Identify fault indications; 4) Describe SONSYM system checks; and 5) Establish communications with fire control.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will introduce the students to the purpose, major components, modes of operation, controls, displays and interfaces of the SONSYM function of the AN/BQQ-10 V7 Sonar Suite. 2) Upon completion of the SONSYM lecture, instructor will provide confirmation questions in order to verify the students have understood the earning objectives.

Resources	References
1) EO 001.01B Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson material; 2) IETM; and 3) NCPM 403.

Notes to Instructor
This is the health monitoring ability for the BQQ-10 Sonar, Ensure the students know this tool for what it is. Conduct confirmation questions to confirm the grasp of the system layout and functions.

EO 001.01 - Perform Operator Checks on Equipment (B0005)

Acoustic Control System

Lesson	Conditions	Sequence
001.01C	Classroom	After 001.01B
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Identify array system components; 2) Identify fault indications; 3) Describe array control system checks; and 4) Describe dental chart hydrophone controls.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will introduce the students to the purpose, major components, modes of operation, controls, displays and interfaces of the Acoustic Controls function AN/BQQ-10 V7 Sonar Suite. 2) Upon completion of the Acoustic Controls lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.01C Presentation; 2) Conformation questions; and 3) Classroom.	1) Lesson material; 2) IETM; and 3) NCPM 403.

Notes to Instructor

It is important for the students to grasp what to look for to indicate a fault on either the system or the hydrophones. Make sure to reiterate the importance of any defect on a sonar, who is to be informed in the event of a defect and the implications of overlooking a defect.

EO 001.01 - Perform Operator Checks on Equipment (B0005)

Set up and perform Operator Checks

Lesson	Conditions	Sequence
001.01D	VCSTT	After EC 001.07
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op

Learning Objectives	Method
1) Perform operator checks on Hydrophone arrays associated with the BQQ-10; 2) Perform system checks on the SONSYM; 3) Perform system checks on the sub-mode displays and processing; and 4) Operate communication equipment (A0010).	EDI

Scenario	The submarine has piped Diving Stations after leaving Halifax harbour. You take your seat and are instructed to perform operator checks on the BQQ-10 and set it up for a dived transit.
Activity Steps	1) Instructor will read the scenario and inform the students when this activity is to be performed. 2) Instructor will demonstrate the initial display and processing set up of all sub-modes, operator checks and long post diving checks on the system. 3) Have each student set up all the display and processing settings for all sub-modes. 4) Instructor will have the students perform operator checks on the SONSYM, Array Control and operability of all sub-modes. 5) Instructor will have each student perform system set up and long post diving checks and verify settings entered are correct. 6) Instructor will provide feedback on any improper procedures or line ups. 7) Instructor will have the students perform this task whenever closing up on the set, taking over a watch or if in doubt of the system operability.

Resources	References
1) VCSTT; 2) Operational BQQ-10 Sonar; and 3) Fire control operator.	1) Lesson Material; 2) IETM; and 3) NCPM 403.

Notes to Instructor

This initial set up of the sonar set is to be conducted every time the students close up in the trainer. The active portion of long post diving checks will be covered in EO 001.03B EDI.

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EC 001.01 - Perform Operator Checks

Scope	Conditions	Sequence
EO 001.01D	Classroom	After completion of 001.01C
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Secret Can/US	MS-PO1 SONAR Op (Subs)

Scenario	You come up on watch and are taking over the set. Verify the settings and sub-systems that need to be verified prior to starting the watch.
Assessment Steps	1) The student is to demonstrate an understanding of the concepts taught in EO 001.01 by answering the scenario-based questions correctly. 2) Student is given the scenario and a questionnaire. Student is expected to answer questions pertaining to the set up and equipment checks covered in EO 001.01 Student has 30 minutes to write the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) EC 001.01 Questionnaire; 2) EC 001.01 Answer Guide; 3) PP&S; and 4) Classroom.	1) Lesson Material 2) NCPM 403

Notes to Instructor
This material is to be used every time the students close up on the set in the VCSTT and subsequently on board when taking over the watch.

Limitations	No references and no assistance allowed. This knowledge will be consolidated during the OJPR.
Impact of Failure	Remedial Training as required and Rewrite.

EO 001.02 - Calculate Acoustic Sensor Range Information (C0115)

Sonar Tactical Decision Aids (STDA)

Lesson	Conditions	Sequence
001.02A	Classroom	After 001.01C
Time	**Classification**	**Instructor**
4 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Explain STDA purpose; 2) Identify capabilities and limitations of STDA; 3) Describe baseline default set-up; and 4) Describe STDA Display modes.	Interactive Lecture

Scenario	Nil
Activity Steps	1) Instructor will introduce the students to the purpose, major components, modes of operation, controls, displays and interfaces of the STDA function AN/BQQ-10 V7 Sonar Suite. 2) Upon completion of the STDA lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.02 Presentation; 2) Conformation Questions; and 3) Classroom.	1) Lesson Material; 2) IETM; and 3) NCPM 403

Notes to Instructor

There is a lot of information covered in this lecture and it will be built on in the OJPR as the students develop briefs and assess oceanographic conditions. Ensure the students have an understanding of this system's capabilities and limitations as a solid foundation.

EO 001.02 - Calculate Acoustic Sensor Range Information (C0115)

Sonar Tactical Decision Aids (STDA)

Lesson	Conditions	Sequence
001.02B	Trainer	After EC 001.07
Time	**Classification**	**Instructor**
4 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Conduct STDA set-up; 2) Input data for range calculation; and 3) Manipulate display settings.	EDI

Scenario	Completed during multiple scenarios throughout the PO 001 trainer phase.
Activity Steps	1) Instructor will demonstrate the initiation and operation of the STDA sub-mode to include but is not limited to; SV/Temp conversion, depth manipulation, map functions, submarine position. 2) The student will be directed to assess the sound velocity profile at intervals throughout the PO 001 trainer phase as required during normal at sea operations.

Resources	References
1) BQQ-10 STDA sub-system enabled; 2) Sound velocity profile; and 3) VCSTT.	1) Lesson Material; 2) IETM; and 3) MILOC Manuel.

Notes to Instructor
This can be done in conjunction with the HFA training as STDA can only be displayed for one student at a time.

EC 001.02 - Calculate Acoustic Sensor Range Information

Scope	Conditions	Sequence
EO 001.02	Classroom	After completion of 001.02A
Time	**Classification**	**Instructor**
1 x 50 Minute Periods	Secret Can/US	MS-PO1, SONAR Op (Subs)

Scenario	You are sailing in the position of a Sound Room Watch keeper. The watch leader directs you to obtain oceanographic information via the STDA sub-system.
Assessment Steps	1) The student is to demonstrate the knowledge of the capabilities and limitations of the STDA sub-system taught in EO 001.02 by answering the knowledge based questions correctly. 2) Student is given the scenario and questionnaire. Student is expected to answer the knowledge based questions correctly. The student has 20 minutes to complete the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) EC 001.02 Questionnaire; 2) EC 001.02 Answer Guide; 3) PP&S; and 4) Classroom.	1) Lesson Material; and 2) IETM.

Notes to Instructor
This material is to be used as part of a sound room watch while at sea or in the VCSTT.

Limitations	No references and no assistance allowed. This knowledge will be consolidated during the OJPR.
Impact of Failure	Remedial Training as required and Rewrite.

EO 001.03 – Operate Active Equipment (T0004)

HF Active Displays and controls

Lesson	Conditions	Sequence
001.03A	Classroom	After 001.02A
Time	**Classification**	**Instructor**
5 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Explain the modes of HFA; 2) Understand the difference between traditional and monopulse processing; 3) Identify capabilities and limitations; 4) Describe HFA tracker operations; 5) Understand the concepts of PUMA; 6) Understand the scribe utility; 7) Explain EMCON policy and impact; and 8) Explain set-up process.	Interactive Lecture

Scenario	Nil
Activity Steps	1) Instructor will introduce the students to the purpose, major components, modes of operation, controls, displays and interfaces of the HFA function AN/BQQ-10 V7 Sonar Suite. 2) Upon completion of the HFA lecture, instructors will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.03A Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) IETM; and 3) NCPM 403.

Notes to Instructor

As sonar ops coming from the fleet, the operators will not have been introduced to some of the features of the HFA. Explain the use of processing filters and normalizers in contact recognition. Pay particular attention to the differences between HFA and the active sonars on the surface ships, when we transmit and why we transmit including the repercussions to transmitting sound into the water as a submarine.

EO 001.03 – Operate Active Equipment (T0004)

Set-up Main Sonar in Active Mode (T0003)

Lesson	Conditions	Sequence
001.03B	VCSTT	After EC 001.07
Time	**Classification**	**Instructor**
2 x 50 Minute Periods (per student)	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Conduct set up of main sonar; 2) Bring contact; 3) Drop contact; 4) Refine tracking; 5) Operate PUMA; 6) Operate MA; 7) Operate ASW; 8) Operate PD; 9) Operate IKA Depth/Fade; and 10) Manipulate settings to achieve optimal performance.	EDI

Scenario	You are in a contact rich environment and the CO suspects that there are contacts close to the submarine. We have to return to periscope depth and you have been ordered to clear the forward arcs with the active sonar. You are in a submarine patrol area and a message comes in with new intel that there are mines in the area and you are to egress immediately. The CO orders the sound room to set the sonar to mine avoidance and sanitize our egress route. You're operating in a CASEX and gain contact on a possible submerged contact the CO wants you to transmit on HFA to verify the fire control solution. A hurricane has passed through the op area and you have been tasked to update the bathymetry in the area prior to an exercise. The UC1 has asked you to conduct a PUMA survey.
Activity Steps	1) Demonstrate the procedures required for each of the Learning Objectives and explain the situations where each operation is to be used. 2) Have the students operate the system in cycle test and input all parameters required for the specific situation.

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	3) Use the preloaded training scenarios built into HFA to familiarize the students with real world representations of active returns from various contacts. 4) Have each student set HFA to MA and conduct mine avoidance. 5) Have each student conduct a puma survey. 6) Return to PD while transmitting HFA and operating the HFA STDA DE steering. 7) Switch pulses between traditional and monopulse to illustrate the different resolutions. 8) Follow checklist for EC 001.03 to ensure all HFA functions are understood. 9) Manipulate settings to show the student that better settings equate to better detections. 10) Promote an active contact via the CAD function to assist in equipment familiarization. 11) On a course of six, one student is to be operating the HFA while two students maintain the passive picture and one student operates the AI&R. The remaining two students are to observe and everyone is to be cycled through each position.

Resources	References
1) VCSTT/TWS; and 2) Operational AN/BQQ-10 system.	1) Lesson material; 2) NCPM 403; and 3) IETM.

Notes to Instructor

Use multiple contacts on a similar course/speed to prolong tracking and assist in learning. Use realistic ranges when creating the scenarios. Due to the limited resources only one student can operate HFA at a time, ensure sufficient time is allotted for the hands on portion. Doubling up with one student on HFA, two students on AI&R and one on STDA will reduce the time required. Main points to reiterate is the process of transmitting in ASW and PD modes with the aid of the STDA thumbnail.

EC 001.03 - Operate Active Equipment

Scope	Conditions	Sequence
EO 001.03	Classroom	After completion of 001.03A
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Secret Can/US	MS-PO1 SONAR Op (Subs)

Scenario	You are sailing in the position of a Sound Room Watch keeper. The watch leader directs you to operate the HFA Sub-system.
Assessment Steps	1) The student is to demonstrate the knowledge of the capabilities and limitations of the HFA sub-system taught in EO 001.03 by answering the knowledge based questions correctly. 2) Student is given the scenario and questionnaire. Student is expected to answer the knowledge based questions correctly. The student has 40 minutes to complete the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) EC 001.03 Questionnaire; 2) EC 001.03 Answer Guide; 3) PP&S; and 4) Classroom.	1) Lesson Material; and 2) IETM.

Notes to Instructor
This material is to be used as part of a sound room watch while at sea or in the VCSTT.

Limitations	No references and no assistance allowed. This knowledge will be consolidated during the OJPR.
Impact of Failure	Remedial Training as required and Rewrite.

EC 001.01-03 – Perform Operator Checks on Equipment

Scope	Conditions	Sequence
EO 001.01-03	VCSTT	After Trainer Phase EDIs
Time	**Classification**	**Instructor**
1 x 50 minute period (per student)	Secret Can/US	MS-PO1 SONAR Op (Subs)

Scenario	You are off the coast of Halifax, open ocean. You are travelling at 57 meters and expect to return to periscope depth on your watch. There is moderate merchant traffic and one warship in the area. You have been tasked to perform setup and operating checks on the AN/BQQ-10 Sonar, display current oceanographic conditions and then transmit on HFA.
Assessment Steps	1) The student is to demonstrate an ability set up the sonar and operate the frame sonar by reacting to direction in a simulated at sea environment. 2) Student is given the scenario and access to the VCSTT. Student is expected to set up and conduct equipment checks covered in EO 001.01. The student will cover the operating procedures for STDA and HFA as covered in EO001.02 and EO 001.03. The Student has 40 Min to complete this test. 3) The instructor will provide direction as per EC 001.01 checklist and may clarify any directions that are not fully understood. 4) Instructor will provide feedback to the individual student, after each scenario, in order to review the tasks and reinforce correct operating procedures.

Resources	References
1) VCSTT 2) CFCD 113 (E) AIG 3) Chronoscope 4) Fire control Operator 5) VCSTT Operator 6) EC 001.01 Checklist	1) Lesson Material 2) NCPM 403 3) IETM

Notes to Instructor

Minimize setup time by building scenarios before commencing the trainer phase. Verify with the trainer staff that the trainer time is booked. Follow and fill the EC 001.01 checklist: one per student and one per run. This information will be used and built upon when performing the duties and responsibilities of a sonar operator on submarines and during the OJPR.

Limitations	No assistance and no references will be provided during the EC.
Impact of Failure	Remedial Training as required and Rewrite.

Item	EC 001.01-03 Operate Main Frame Sonar	SAT	UNSAT	Remarks
1. Perform closing up checks to include.				
Perform a communications check.		☐	☐	
Check cursor response on all sensors.		☐	☐	
Verify audio on all sensors.		☐	☐	
Check the status of arrays.		☐	☐	
Check ship's and array heading follows actual ship's heading.		☐	☐	
Ensure navigation data is correct.		☐	☐	
Verify that the SONSYM is operational.		☐	☐	
Input correct display settings to include screen colour, baffle indicators, ship's heading marker, water falling sierra numbers and sonar logger is operational for following displays: 1. CBB; 2. Multi Hull; 3. PNB; 4. AI&R; and 5. HFA.		☐	☐	
Ensure acoustic information is updating.		☐	☐	
Set broadband vertical beam width to wide.		☐	☐	
Set narrowband normalizers to narrow.		☐	☐	
Check that server temperature levels are within tolerances.		☐	☐	
2. Start STDA and select the following.				
Set submarine position to in situ.		☐	☐	
Switch map display between 2D and 3D.		☐	☐	
Manipulate sound velocity display to include depth and unit of measurements.		☐	☐	
3. Operate HFA.				
Select HFA Mode.		☐	☐	
Select Cycle Test.		☐	☐	
Initiate single transmission in Cycle Test.		☐	☐	

Item	EC 001.01-03 Operate Main Frame Sonar	SAT	UNSAT	Remarks
Adjust Range scale.		☐	☐	
Adjust Bearing coverage.		☐	☐	
Switch between sub-modes to include: 1. ASW; 2. IKA – Depth; 3. IKA – Fade; 4. Mine Avoidance; 5. PD; and 6. PUMA.		☐	☐	
Switch between display options to include: 1. PPI Intensity; 2. PBB/PPI; 3. B-Scan Single; 4. B-Scan Double; 5. B-Scan Triple; and 6. PPI Depth.		☐	☐	
Track a contact actively. (ASW/PD)		☐	☐	
Select Pulse type.		☐	☐	
Update active contact. (Track Expand Window)		☐	☐	
Enable/Disable CAD.		☐	☐	
Set Standoff Range. (Mine Avoidance)		☐	☐	
Set Safety Corridor. (Mine Avoidance)		☐	☐	
Operate STDA Thumbnail. (PD/ASW)		☐	☐	
Correlate Active contact with HFA PBB Wrap Around.		☐	☐	
Interpret Depth. (IKA/PPI-Depth)		☐	☐	
Toggle CAD overlays.		☐	☐	
Toggle External Tracker Repeaters.		☐	☐	
Switch frequency via Pre-sets. (Processing)		☐	☐	
Select processing type. (Monopulse/2D Single/2D Triple)		☐	☐	
Identify Primary, Surface and Bottom Reject contacts. (Monopulse/B-SCSAB)		☐	☐	
Set Depression Elevation.		☐	☐	
Use Beam Normalizers to suppress Broadband spokes. (PPI)		☐	☐	

Item	EC 001.01-03 Operate Main Frame Sonar	SAT	UNSAT	Remarks
Match Sierra number to Fire Control Number.		☐	☐	

Score: **/ 39**

The trainee must achieve 30/39 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

NTDC(A) Approval – 02/03/2021
NPTG HQ Approval -10/05/2019
Version No: 5.1
MITE Crse ID : 109272/109273

EO 001.04 - Operate Passive Equipment(s) (T0002)

CF Promotion (PBB and PNB)

Lesson	Conditions	Sequence
001.04 A	Classroom	After EC 001.01Written
Time	**Classification**	**Instructor**
3 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Identify capabilities and limitations of tracking system; 2) Assign manual points; 3) Perform CF trace editing; 4) Describe process to track contact; and 5) Explain contact follower management process.	Interactive Lecture / Video

Scenario	Nil.
Activity Steps	1) Start the lesson with the lecture to provide base knowledge and then show the students the video on CF promotion. Encourage questions throughout the video and clarify anything covered that is not fully understood. 2) Once the video on CF promotion is finished, instructor will show the video on trace editing. 3) On completion of the CF trace editing video, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.04A Presentation; 2) Confirmation Questions; 3) Classroom; 4) CF promotion video; and 5) CF trace editing video.	1) Lesson Material; 2) IETM; and 3) NCPM 403.

Notes to Instructor
CF promotion skills are used every day on board, ensure students have an understanding of the differences between precision trackers and contact followers.

EO 001.04 - Operate Passive Equipment(s) (T0002)

Track Broadband (T0017) – PBB Common Broadband

Lesson	Conditions	Sequence
001.04 B	Classroom	After 001.04 A
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Describe purpose of common broadband system display; 2) Familiarization of menu bars, cursors and fast action buttons; 3) Become fluent in trace overlay colour schemes; 4) Describe capabilities and limitations of common broadband system; 5) Explain initial set-up of common broadband system; 6) Initiate a scissor gram; and 7) Initiate a linked lofar gram.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 001.04B lecture on CBB operations, capabilities and limitations. 2) On completion of the CBB lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.04B Presentation; 2) Confirmation Questions; and 3) Classroom;	1) Lesson Material; 2) IETM; and 3) NCPM 403.

Notes to Instructor
Ensure the students are taking notes on the capabilities of the set as it is a lot of material to cover.

EO 001.04 - Operate Passive Equipment(s) (T0002)

Track Broadband (T0017) - PBB Multi-Hull

Lesson	Conditions	Sequence
001.04 C	Classroom	After 001.04 B
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Describe purpose of multi-hull; 2) Be familiar with sensor assignment control menus; 3) Describe capabilities and limitations of multi-hull; and 4) Explain initial set-up of multi-hull.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 001.04C lecture on Multi-Hull operations, capabilities and limitations. 2) On completion of the Multi-Hull lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.04C Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) IETM; and 3) NCPM 403.

Notes to Instructor
Reiterate to the students that this is the “Safety of the Submarine” screen and is to be kept visible on one of the operators screen at all times.

EO 001.04 - Operate Passive Equipment(s) (T0002)

Track Narrowband (T0018)

Lesson	Conditions	Sequence
001.04 D	Classroom	After 001.04C
Time	**Classification**	**Instructor**
4 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Describe purpose of the PNB sub-mode; 2) Describe capabilities and limitations of the PNB sub-mode; 3) Explain functions of PNB; 4) Understand beam OR'ing schemes; 5) Understand the mechanics of steering PNB search grams via the bearing sector overlay; 6) Explain the process used in the classification of contacts; 7) Explain the optimum set up of PNB; 8) Describe the tracking process with vertical grams; 9) Understand sector colour overlays' indication of bearing movement; 10) Understand gram selection vs displayed history; 11) Understand the use of PNB hybrid contact followers; 12) Understand the use of multiple cursor types; and 13) Understand the conduct of a methodical search routine.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 001.04D lecture on PNB operations, capabilities and limitations. 2) On completion of the PNB lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.04D Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) IETM; and 3) NCPM 403.

Notes to Instructor
Ensure the students are aware that there are many different ways to access the PNB sub modes from either the main menu or through the passive sub-systems. (LLG SCG etc.)

EO 001.04 - Operate Passive Equipment(s) (T0002)

Track Narrowband (T0018) – Class FASTGRAM

Lesson	Conditions	Sequence
001.04 E	Classroom	After 001.04D
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Describe purpose of FASTGRAM; 2) Describe capabilities and limitations of FASTGRAM; 3) Explain functions of FASTGRAM; 4) Explain classification of FASTGRAM; 5) Explain set up of FASTGRAM; and 6) Understand the process of conducting a ping steal ranging.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 001.04E lecture on Fastgram operations, capabilities and limitations. 2) On completion of the Fastgram lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.04E Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) IETM; and 3) NCPM 403.

Notes to Instructor
Make a note to the students that this sub-mode is very similar to AI&R and has a lot of the same characteristics.

EO 001.04 - Operate Passive Equipment(s) (T0002)

Track Narrowband (T0018) – DEMON and Interference Pattern Features

Lesson	Conditions	Sequence
001.04 F	Classroom	After 001.04E
Time	**Classification**	**Instructor**
2 x 50 Minute Period	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Describe purpose of DEMON; 2) Describe capabilities and limitations of DEMON; 3) Explain the functions, feature indicators, alerts and annotation of DEMON; 4) Explain the process of contact classification using DEMON; 5) Explain set up of DEMON; 6) Understand the process used to enter TPK for speed calculations; 7) Understand the interference pattern indicators; 8) Understand the interaction between DEMON input and the Contact information readout; and 9) Describe the feature CF suppression time controls.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 001.04F lecture on DEMON and interference pattern operations, capabilities and limitations. 2) On completion of the DEMON and interference patterns lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.04F Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) IETM; and 3) NCPM 403.

Notes to Instructor
Ensure that the students understand that this sub-mode, in no way, replaces the advantages of a PAL assessment of the contact, as there are aural characteristics that provide valuable tactical information in real time.

EO 001.04 - Operate Passive Equipment(s) (T0002)

Track Narrowband (T0018) – Integrated Towed Array

Lesson	Conditions	Sequence
001.04 G	Classroom	After 001.04F
Time	**Classification**	**Instructor**
4 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Describe purpose of Integrated Towed Array; 2) Describe capabilities and limitations of Integrated Towed Array; 3) Explain functions of Integrated Towed Array; 4) Explain classification with Integrated Towed Array; 5) Explain set up of Integrated Towed Array; 6) Describe tracking process of Integrated Towed Array; 7) Describe the physical composition of the Towed Array; and 8) Explain the process for solving for ambiguity.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 001.04F lecture on Towed Array operations, capabilities and limitations. 2) On completion of the Towed Array lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.04G Presentation; 2) Confirmation Question; and 3) Classroom.	1) Lesson Material; 2) IETM; and 3) NCPM 403.

Notes to Instructor
Nil.

EO 001.04 – Operate Passive Equipment(s) (T0002)

Track Acoustic Intercept and Ranging (AI&R) (T0016)

Lesson	Conditions	Sequence
001.04 H	Classroom	After 001.04G
Time	**Classification**	**Instructor**
3 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Describe purpose of AI&R; 2) Describe the inputs of AI&R; 3) Explain the display options of AI&R; 4) Describe capabilities and limitations of AI&R; 5) Explain functions of AI&R; 6) Understand Cursor options and cursor operation; 7) Describe gram controls and audio controls; 8) Describe alert bands, detector controls and SCANR; 9) Explain classification with AI&R; 10) Explain set up of AI&R; and 11) Describe tracking process of AI&R.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 001.04I lecture on AI&R operations, capabilities and limitations. 2) On completion of the AI&R lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.04I Presentation; 2) Confirmation Questions; and 3) Classroom	1) Lesson Material; 2) ADAC Pub 4A 3) IETM; and 4) NCPM 403.

Notes to Instructor
Nil.

EO 001.04 - Operate Passive Equipment(s) (T0002)

Track Narrowband (T0018) – 2046 Analyzer

Lesson	Conditions	Sequence
001.04 I	Classroom	After 001.04H
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Describe purpose of 2046; 2) Describe capabilities and limitations of 2046; 3) Explain fault identification of 2046; 4) Describe the components of the 2046; 5) Describe the input sources of the 2046; 6) Explain set up of 2046; 7) Describe tracking process and functions of 2046; 8) Explain the process for solving for ambiguity; 9) Describe the display format of the 2046; 10) Describe the function of the different displays; 11) Understand the use of combined profiles during tracking; 12) Describe the voice procedures of the 2046; 13) Describe the closing up procedure of the 2046; 14) Understand vernier resolution calculation; and 15) Understand the procedure for close quarters contacts on 2046.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 001.04H lecture on 2046 operations, capabilities and limitations. 2) On completion of the 2046 lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.04H Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) IETM; and 3) NCPM 403.

Notes to Instructor
This lecture is to be given to the students until the 2046 has been removed from all operational submarines. At that point it is superseded by 001.04G.

EO 001.04 - Operate Passive Equipment(s) (T0002)

Track Acoustic Contacts (C0070)

Lesson	Conditions	Sequence
001.04J	VCSTT	After EC 001.07
Time	**Classification**	**Instructor**
40 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op

Learning Objectives	Method
1) Operate cursors in all modes; 2) Perform Torpedo drills; 3) Perform return to periscope drills; 4) Perform close quarters drills to include fishing vessel avoidance; 5) Perform a stern arc clearance; 6) Perform an emergency go deep drill; 7) Set up the CBB display; 8) Conduct a passive search in CBB; 9) Identify contacts using CBB; 10) Promote contacts using CBB; 11) Edit contacts using CBB; 12) Send a contact to linked lofar gram; 13) Send a contact to a scissor gram; 14) Set up the display for MHD; 15) Conduct a passive search in MHD; 16) Identify contacts using MHD(BOW/PRS); 17) Promote contacts using MHD(BOW/PRS); 18) Edit contacts using MHD; 19) Set up the PNB display; 20) Conduct a narrowband search; 21) Identify contacts using PNB; 22) Promote contacts using PNB; 23) Track contacts using PNB vertical grams; 24) Identify target maneuvers using PNB; 25) Classify contacts using PNB; 26) Calculate bearing movement; 27) Set hybrid contact follower frequency and bandwidth; 28) Send classification to fire control and sonar logger; 29) Set FASTGRAM sensor; 30) Set FASTGRAM center frequency and bandwidth; 31) Calculate ping steal range; 32) Set up DEMON display; 33) Perform a DEMON search; 34) Classify a contact using DEMON;	EDI

35) Calculate TPK using DEMON(LLG); 36) Interrogate a feature CF; 37) Identify an interference pattern; 38) Set Feature CF suppression time; 39) Set up AI&R display; 40) Set threat and torpedo alert bands; 41) Identify a contact on integrated towed array; 42) Track a contact on integrated towed array; 43) Classify a contact on integrated towed array; 44) Perform a triangulation range on integrated towed array; 45) Set display settings for integrated towed array; 46) Solve for ambiguity on integrated towed array; 47) Set audio source on AI&R; 48) Set input sources for AI&R; 49) Identify contacts using AI&R; 50) Promote contacts using AI&R; 51) Classify contacts using AI&R; 52) Report transmissions using AI&R; 53) Perform a ping steal range using AI&R; 54) Set up 2046 screen sequencing; 55) Set Vernier center frequency, bandwidth and resolution; 56) Identify a contact on 2046; 57) Track a contact on 2046; 58) Solve for ambiguity on 2046; 59) Track a contact in combined profiles 2046; 60) Identify a contact on towed broadband 2046; 61) Correlate towed broadband (2046) with Bow/PRS; and 62) Send broadband contact to fire control on 2046.	

Scenario	You are at sea, in international waters. There is a task group in your area. Your unit has been tasked to close within visual distance and conduct surveillance. Video, ACINT, ELINT and all other information is to be forwarded to Halifax for analysis as soon as you break the engagement. You are at the BQQ-10 8002 console and are tasked between PBB, PNB, AI&R, NM and the Towed Array. The watch leader has briefed you on the threat frequencies, active sonars, torpedoes and TPKs of the enemy units.
Activity Steps	1) Initially have the student, under direction, set up all passive sub-systems of BQQ-10 and 2046. Ensure that the threat frequencies are covered by hybrid contact followers and 2046 verniers. Ensure active alert bands are set to threat and torpedo frequencies.

	2) Run through a slow time scenario to familiarize the students with the functions of the passive sub-system. Follow the EC001.02 passive sub-systems checklist to ensure all learning objectives are met. 3) Once the students have a grasp of the basic commands and actions, take them through a more dynamic contact situation to include crossing contacts and contact zigs. 4) Once the students display an understanding of intermediate contact management skills (Dynamic contacts, CF editing, contact crossings), initiate scenarios that include return to periscope depth and stern arc clearance. In order to practice torpedo drills and close quarters drills, incorporate this into each scenario and have each student perform the drills independently. 5) Adaptive scenarios are to be used near the end of the trainer phase in order to address any deficiency identified by the instructor during the developmental runs.

Resources	References
1) VCSTT; 2) Operational AN/BQQ-10 Sonar System; and 3) Fire control operator.	1) NCPM 403; 2) CFCD 113 (D) Vol 2 AIG; 3) IETM; and 4) ADAC Pub 4A.

Notes to Instructor

Recommend one contact per student to avoid confusion in the beginning. Have the contacts on a similar course and speed of the submarine to prolong in contact time and give the students time to manipulate tracking. This trainer phase can be combined with the rest of the material taught in PO 001.

EO 001.04 - Operate Passive Equipment(s) (T0002)

Capabilities and Limitations review

Lesson	Conditions	Sequence
001.04 K	Classroom	After EO 001.04 J
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Describe capabilities and limitations (EO 001.01-04).	Self-study

Scenario	Nil.
Activity Steps	1) Start the lesson with explaining that this time is provided to them to help retain the vast quantity of knowledge required to safely operate the main frame sonar. They have the length of this lesson to review their notes from EO 001.01 to EO 001.04, quiz themselves and ask further questions if needed. The sounds level should remain low at all time. 2) This period will be followed by the capabilities and limitations written EC.

Resources	References
1) Assistance; 2) Supervision; 3) Classroom; and 4) References.	1) Lesson Material; and 2) NCPM 403.

Notes to Instructor
Ensure students understanding by walking the classroom. Interacting with students based on verbal and non-verbal requests to answer questions and provide further information.

EO 001.04 - Operate Passive Equipment(s) (T0002)

Capabilities and Limitations review

Lesson	Conditions	Sequence
001.04 L	Classroom	After EO 001.07 B
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Describe capabilities and limitations (EO 001.01-04)	Self-study

Scenario	Nil
Activity Steps	1) Start the lesson with explaining that this time is provided to them to help retain the vast quantity of knowledge required to safely operate the main frame sonar. They have the length of this lesson to review their notes from EO 001.01 to EO 001.04, quiz themselves and ask further questions if needed. The sounds level should remain low at all time. 2) This period will be followed by the written capabilities and limitations written PC.

Resources	References
1) Assistance; 2) Supervision; 3) Classroom; and 4) References.	1) Lesson Material; and 2) NCPM 403.

Notes to Instructor
Ensure students understanding by walking the classroom. Interacting with students based on verbal and non-verbal requests to answer questions and provide further information.

EC 001.04 - Operate Passive Equipment(s) (T0002)

Scope	Conditions	Sequence
EO 001.04	Classroom	After completion of 001.04I
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret Can/US	MS-PO1, SONAR Op (Subs)

Scenario	You are sailing in the position of a Sound Room Watch keeper. The watch leader directs you to operate the passive sub-system to include PBB, PNB, NM and AI&R.
Assessment Steps	1) The student is to demonstrate the knowledge of the capabilities and limitations of the PBB, PNB, NM and AI&R sub-systems taught in EO 001.04 by answering the knowledge based questions correctly. 2) Student is given the scenario and questionnaire. Student is expected to answer the knowledge based questions correctly. The student has 40 minutes to complete the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) EC 001.04 Questionnaire; 2) EC 001.04 Answer Guide; 3) PP&S; and 4) Classroom.	1) Lesson Material; and 2) IETM.

Notes to Instructor
This material is to be used as part of a sound room watch while at sea or in the VCSTT.

Limitations	No references and no assistance allowed. This knowledge will be consolidated during the OJPR.
Impact of Failure	Remedial Training as required and Rewrite.

EO 001.05 - Conduct Self Noise Monitoring (T0019)

Noise Monitoring (NM)

Lesson	Conditions	Sequence
001.05 A	Classroom	After 001.04 I
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Explain purpose of the Noise Monitoring sub-system; 2) Identify components of Noise Monitoring; 3) Explain capabilities and limitations of Noise Monitoring; 4) Explain functions of Noise Monitoring; and 5) Identify artifact of Noise Monitoring.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will introduce the students to the purpose, major components, modes of operation, controls, displays and interfaces of the NM function AN/BQQ-10 V7 Sonar Suite. 2) Upon completion of the NM lecture, instructor will provide questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.05A Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) IETM; and 3) NCPM 403.

Notes to Instructor

The information covered in this lecture will be built on in the OJPR. As students learn to incorporate the process of Self noise monitoring as part of normal watch rotation. Ensure the students have an understanding of this system's capabilities and limitations as a solid foundation.

EO 001.05 - Conduct Self Noise Monitoring (T0019)

Carry Out Noise Monitoring (NM) Process

Lesson	Conditions	Sequence
001.05 B	VCSTT	After EC 001.07
Time	**Classification**	**Instructor**
1 x 50 Minute Period (per Student)	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Operate the Noise Monitoring sub-system.	EDI

Scenario	At sea off the coast, you are tasked to investigate a possible artefact on the PRS MA array.
Activity Steps	1) Have the student select the NM sub-system. 2) Have the student select the correct array/hydrophone/stave and investigate the report of a defect in single sensor analysis processing. 3) Demonstrate the set up and monitoring of the self-noise detection displays. 4) Have each student go through the process of localizing a defect using the SDT hydrophones and self-noise detection sub-menu. 5) This process is to be repeated throughout the trainer phase to ensure the student retains the operating skills.

Resources	References
1) VCSTT/TWS; and 2) Operational AN/BQQ-10 system.	1) Lesson Material; 2) IETM; and 3) NCPM 403.

Notes to Instructor

The information covered in this lecture will be built on in the OJPR. As students learn to incorporate the process of Self noise monitoring as part of normal watch rotation. Ensure the students have an understanding of this system's capabilities and limitations as a solid foundation.

EC 001.05 – Conduct Self Noise Monitoring

Scope	Conditions	Sequence
EO 001.05	Classroom	After completion of EO 001.05A
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Secret Can/US	MS-PO1 SONAR Op (Subs)

Scenario	You are sailing in the position of a Sound Room Watch keeper. The watch leader directs you to operate the self-noise monitoring sub-system.
Assessment Steps	1) The student is to demonstrate the knowledge of the capabilities and limitations of the Self Noise Monitoring sub-systems taught in EO 001.05 by answering the knowledge based questions correctly. 2) Student is given the scenario and questionnaire. Student is expected to answer the knowledge based questions correctly. The student has 40 minutes to complete the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) EC 001.05 Questionnaire 2) EC 001.05 Answer Guide 3) PP&S 4) Classroom	1) Lesson Material 2) IETM

Notes to Instructor
This material is to be used at sea as part of a sound room watch while at sea or in the VCSTT.

Limitations	No references and no assistance allowed. This knowledge will be consolidated during the OJPR.
Impact of Failure	Remedial Training as required and Rewrite.

EO 001.06 – Record Acoustic Intelligence (ACINT) Data (T0020)

Digital Data Collection System (DDCS)

Lesson	Conditions	Sequence
001.06A	Classroom	After 001.05A
Time	**Classification**	**Instructor**
3 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Explain purpose of DDCS; 2) Explain components of DDCS; 3) Explain capabilities and limitations of DDCS; 4) Explain set up of DDCS; 5) Explain DDCS controls, displays and indication; 6) Describe process of DDCS; 7) Describe the inputs of DDCS; 8) Explain the process used in archiving screen captures to disk; and 9) Describe the media labeling process.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will introduce the students to the purpose, major components, modes of operation, controls, displays and interfaces of the DDCS function AN/BQQ-10 V7 Sonar Suite. 2) Upon completion of the DDCS lecture, instructor will provide questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.06A Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) IETM; and 3) CSRM.

Notes to Instructor
Current policy is to conduct event recording, ensure the students know to check with their UC1 when proceeding to sea to verify individual submarine policy.

EO 001.06 – Record Acoustic Intelligence (ACINT) Data (T0020)

Sonar Logger

Lesson	Conditions	Sequence
001.06B	Classroom	After 001.06A
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Explain purpose of Sonar Logger; 2) Explain capabilities and limitations of Sonar Logger; 3) Describe content input into Sonar Logger; 4) Describe annotation process into Sonar Logger; 5) Explain log recall of Sonar Logger; 6) Describe the controls used to archive to portable media; and 7) Describe information flow of Sonar Logger.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will introduce the students to the purpose, major components, modes of operation, controls, displays and interfaces of the Sonar Logger function AN/BQQ-10 V7 Sonar Suite. 2) Upon completion of the Sonar Logger lecture, instructor will provide questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.06B Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) IETM; and 3) CSRM

Notes to Instructor
Make sure to reiterate to the students that as the AN/BQQ-10 Sonar suite is classified Secret Can/US, any material they take from that system for any reason is to be treated with the same classification.

EO 001.06 – Record Acoustic Intelligence (ACINT) Data (T0020)

Operate Digital Data Collection System (T0022)

Lesson	Conditions	Sequence
001.06C	VCSTT/TWS	After EC 001.07
Time	**Classification**	**Instructor**
1 x 50 Minute Period (per Student)	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Set hydrophone input sources; 2) Perform event and continuous recording; 3) Copy a tape; 4) Write a screen capture to disk; and 5) Lease a drive to scribe recorder.	EDI

Scenario	You're about to start an exercise and are tasked to ensure the BQQ-10 DDCS system is recording.
Activity Steps	1) Have the students open the DDCS controls and input hydrophone sources. 2) Set DDCS Hyd and AEMP recording to cover the exercise time frame via both event and continuous recording methods. 3) Have the student recall a screen capture and write it to disk. 4) Have the student lease a drive to the scribe recorder. 5) Perform a tape duplication.

Resources	References
1) VCSTT/TWS; 2) LTO4 tapes; 3) Blank DVDR; and 4) Operational BQQ-10 Sonar.	1) Lesson material; 2) NCPM 403; and 3) IETM.

Notes to Instructor
Resources for LTO4 tapes are slim, ensure you can use a tape from the TWS or contact ADAC (A) if you need.

EO 001.06 – Record Acoustic Intelligence (ACINT) Data (T0020)

Operate Sonar Logger (T0021)

Lesson	Conditions	Sequence
001.06D	VCSTT	After EC 001.07
Time	**Classification**	**Instructor**
1 x 50 Minute Period (per Student)	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Input content into Sonar Logger; 2) Annotate data from Sonar Logger; and 3) Recall of Sonar Logger data.	EDI

Scenario	The UC1 has tasked you with entering information into the log, on your watch.
Activity Steps	1) While conducting the developmental trainer runs, designate one student per run as logger. 2) Brief the logger as to his/her responsibilities with regards to frequency of logging and content. 3) Have them input pertinent information for contacts into the sonar logger. 4) The instructor will direct the logger to recall information pertaining to a specific contact or specific time.

Resources	References
1) VCSTT; and 2) Operational BQQ-10.	1) NCPM 403; 2) CSRM; 3) IETM; and 4) Lesson material.

Notes to Instructor
The authority to dictate recording policy and logging practices sits with the XO (Records Officer). The policy on the boat will change based on the exercise. The XO should provide a brief to all record keepers prior to serial start.

EC 001.06 – Recording Acoustic Intelligence (ACINT) Data

Scope	Conditions	Sequence
EO 001.06	Classroom	After completion of EO 001.06B
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Secret Can/US	MS-PO1, SONAR Op (Subs)

Scenario	You are sailing in the position of a Sound Room Watch keeper. The watch leader directs you to operate DDCS sub-system.
Assessment Steps	1) The student is to demonstrate the knowledge of the capabilities and limitations of the Digital Data Collection System taught in EO 001.06 by answering the knowledge based questions correctly. 2) Student is given the scenario and a questionnaire. Student is expected to answer the knowledge based questions correctly. The student has 40 minutes to complete the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) EC 001.06 Questionnaire; 2) EC 001.06 Answer Guide; 3) PP&S; and 4) Classroom.	1) Lesson Material; and 2) IETM.

Notes to Instructor
This material is to be used as part of a sound room watch while at sea or in the VCSTT.

Limitations	No references and no assistance allowed. This knowledge will be consolidated during the OJPR.
Impact of Failure	Remedial Training as required and Rewrite.

EC 001.04-06 – Operate Passive, NM and DDCS Sub-Systems

Scope	Conditions	Sequence
EO 001.04-06	VCSTT	After Trainer Phase EDIs
Time	**Classification**	**Instructor**
2 x 50 minute periods (per Student)	Secret Can/US	MS-PO1, SONAR Op (Subs)

Scenario	You are off the coast of Halifax, open ocean. You are travelling at 57 meters and expect to return to periscope depth on your watch. Oceanographic conditions are as briefed in EC 001.07. There is moderate merchant traffic and one warship in the area.
Assessment Steps	1) The student is to demonstrate an ability operate the main frame sonar by reacting to direction in a simulated at sea environment. 2) Student is given the scenario and access to the VCSTT. Student is expected operate the Passive, NM and DDCS sub-systems as directed by the instructor. Student has 40 Minutes to complete this test. 3) The instructor will provide direction as per EC 001.02 checklist and may clarify any directions that are not fully understood. 4) Instructor will provide feedback to the individual student, after each scenario, in order to review the tasks and reinforce correct operating procedures.

Resources	References
1) VCSTT; 2) CFCD 113 (E) AIG; 3) Chronoscope; 4) Fire control Operator; 5) VCSTT Operator; and 6) EC 001.02 Checklist.	1) Lesson Material; 2) NCPM 403; and 3) IETM.

Notes to Instructor

Minimize setup time by building scenarios before commencing the trainer. Verify with the trainer staff that the trainer time is booked. Follow and fill the EC 001.02 checklist: one per student and one per run. This information will be used and built upon when performing the duties and responsibilities of a sonar operator on submarines and during the OJPR.

Limitations	No assistance will be provided during the EC.
Impact of Failure	Remedial Training as required and Rewrite.

Item	EC 001.02 OPERATE PASSIVE, NM AND DDCS SUB-SYSTEMS	SAT	UNSAT	Remarks
1. Operate passive sub-systems.				
-PBB-				
Initiate contact: 1. Auto Tracker Bow/PRS; and 2. Manual Tracker Bow/PRS.		☐	☐	
Initiate Torpedo drills **(CRITICAL)**		☐	☐	
Switch between Auto and Manual tracking		☐	☐	
Determine range from contact and assess tabs.		☐	☐	
Use track windows to refine PRS solution. (MHD)		☐	☐	
Calculate Relative Bearing.		☐	☐	
Switch between PRS FM and MA arrays.		☐	☐	
Enable/disable Global Overlays.		☐	☐	
Initiate a Scissor Gram. (CBB)		☐	☐	
Operate Cursors to include: 1. Crosshair; 2. Delta; and 3. Annotate gram.		☐	☐	
Switch between multiple arrays. (CBB/MHD)		☐	☐	
Carry out voice procedures IAW NCPM 403 for: 1. New Contact; 2. Amplify Contact; and 3. Lost Contact.		☐	☐	
Carry out Close Quarters Drill **(CRITICAL)**		☐	☐	
Switch between frequency bandwidths. (CBB/MHD)		☐	☐	
Locate and relay Non Acoustic Data to include: 1. Date/Time; 2. Heading; 3. Depth; 4. Speed; 5. Pitch/Roll; 6. Keel to Bottom Depth;		☐	☐	

Item	EC 001.02 OPERATE PASSIVE, NM AND DDCS SUB-SYSTEMS	SAT	UNSAT	Remarks
	7. Sound Velocity; and 8. Array Heading (if fitted).			
	Toggle specific overlays. (CBB/MHD)	☐	☐	
	Toggle contact symbols.	☐	☐	
	Interrogate Feature CF.	☐	☐	
	Recall and store display configurations.	☐	☐	
	Set Hybrid Contact Follower center frequency and bandwidth.	☐	☐	
	Manipulate Audio Filters.	☐	☐	
	Set Active Rejection band filter center frequency and bandwidth.	☐	☐	
	Rig for black.	☐	☐	
	Access calculator.	☐	☐	
	Measure bearing.	☐	☐	
	Measure Bearing Rate. (Bow/PRS)	☐	☐	
	Determine Intensity.	☐	☐	
	Edit contact history via CF stitching functions.	☐	☐	
	Interrogate acoustic energy to identify contacts.	☐	☐	
	Conduct Stern arc clearance (MHD)	☐	☐	
	Select and deselect Grams displayed. (CBB)	☐	☐	
	Adjust Update Rate. (CBB/MHD)	☐	☐	
	Switch Vertical Beam Width. (CBB/MHD)	☐	☐	
	Conduct Triangulation Ranging.	☐	☐	
	Drill Down frequency bandwidths. (CBB)	☐	☐	
	Recognize target Zig.	☐	☐	
-PNB-				
	Track a NB Contact via Vertical Grams.	☐	☐	
	Switch formats between: 1. Full Frequency; 2. Paged; 3. Demon; 4. Linked LOFAR Gram; 5. Scissor Gram; and	☐	☐	

Item	EC 001.02 OPERATE PASSIVE, NM AND DDCS SUB-SYSTEMS	SAT	UNSAT	Remarks
	6. Fastgram.			
	Set input source. (Fastgram)	☐	☐	
	Set frequency BW and center frequency. (Fastgram)	☐	☐	
	Select and center ALI display. (Fastgram)	☐	☐	
	Determine B/S configuration and Rev count.	☐	☐	
	Send Classification information to Log.	☐	☐	
	Switch between contacts on Linked Lofar Gram.	☐	☐	
	Switch input source between multiple hydrophone arrays.	☐	☐	
	Manipulate cursor style and colour.	☐	☐	
	Identify Hybrid Contact Follower bandwidth and center frequency.	☐	☐	
	Interpret cursor read our area to include: 1. Frequency; 2. Bearing; 3. Time; 4. Δ Time of event; and 5. Δ Frequency.	☐	☐	
	Use cursers to include: 1. Crosshair; 2. Harmonic; 3. Comb; and 4. Delta.	☐	☐	
	Change update rate.	☐	☐	
	Verify Normalizers are set to narrow.	☐	☐	
	Select Demon bandwidth.	☐	☐	
	Update Vertical Gram position.	☐	☐	
	Scroll through narrowband bearings with: 1. Paged; 2. Full Frequency Single; 3. Demon; 4. 8 Ored; and 5. 16 Ored.	☐	☐	
	Manipulate the Bearing tape to reposition Analysis Grams and Vertical Grams.	☐	☐	

Item	EC 001.02 OPERATE PASSIVE, NM AND DDCS SUB-SYSTEMS	SAT	UNSAT	Remarks
Correlate BB and NB contacts.		☐	☐	
Change Vertical Beamwidth.		☐	☐	
Initiate Torpedo drills **(CRITICAL)**		☐	☐	
Drop Contact.		☐	☐	
Regain Contact.		☐	☐	
Recognize NB indication of target Zig.		☐	☐	
Identify bearing movement based on coloured overlays.		☐	☐	
-AI&R-				
Select AI&R display.		☐	☐	
Switch between display formats.		☐	☐	
Switch between true and relative bearing.		☐	☐	
Operate cursors to include: 1. Scroll; 2. Crosshair; 3. Delta; and 4. Zoom.		☐	☐	
Switch between input sources.		☐	☐	
Set and select Auxiliary inputs.		☐	☐	
Select audio source hydrophones.		☐	☐	
Recognize hydrophone status via indicators.		☐	☐	
Relay cursor readout information.		☐	☐	
Cycle quick lineup buttons to include: 1. Normal; 2. PD; and 3. Torp.		☐	☐	
Switch between Bearing/Frequency displays.		☐	☐	
Change update rates.		☐	☐	
Select track controls to include: 1. BTH; 2. SPLTH; 3. FBI; and 4. FTH.		☐	☐	
Measure pulse length.		☐	☐	
Conduct Ping Stolen Range.		☐	☐	

Item	EC 001.02 OPERATE PASSIVE, NM AND DDCS SUB-SYSTEMS	SAT	UNSAT	Remarks
Adjust Frequency bandwidth and center frequency.		☐	☐	
Switch between Contact and Event history.		☐	☐	
Promote active emissions to a Papa tracker.		☐	☐	
Promote active emissions to a Sierra tracker.		☐	☐	
Clear History.		☐	☐	
Set an Alert Band.		☐	☐	
Configure contact association for bearing and frequency.		☐	☐	
Set detector sensitivity.		☐	☐	
Set Own Ship Muting on/off.		☐	☐	
Configure Scanner.		☐	☐	
Set Quick Line Up to WL/UC1 settings.		☐	☐	
2. Operate noise monitoring equipment.				
Select noise monitoring.		☐	☐	
Select hydrophone/array.		☐	☐	
Manipulate frequency bandwidth and center frequency.		☐	☐	
Select Self noise monitoring.		☐	☐	
Cycle through sensors.		☐	☐	
3. Record acoustic intelligence data.				
Open the DDCS window.		☐	☐	
Configure Hydrophone Inputs.		☐	☐	
Configure Sail recorder input.		☐	☐	
Assign a tape to: 1. DDCS Hyd; 2. AEMP; and 3. Tools.		☐	☐	
Open Recording Tab.		☐	☐	
Initiate event recording/continuous recording.		☐	☐	
Copy a tape via Media utilities.		☐	☐	
Take a Screen capture via Menu bar and Alert Bar.		☐	☐	

Item	EC 001.02 OPERATE PASSIVE, NM AND DDCS SUB-SYSTEMS	SAT	UNSAT	Remarks
	Recall the screen capture via the Screen Capture Viewer.	☐	☐	
	Manipulate the Zoom function in the Screen Capture Viewer.	☐	☐	
	Burn Screen Capture to disk.	☐	☐	
	Annotate an event with the Sonar Logger.	☐	☐	
	Annotate a contact with the Sonar Logger.	☐	☐	
	Start the Scribe recorder.	☐	☐	
	Lease an LTO4 tape to the Scribe Recorder and record HFA.	☐	☐	
4. Assess oceanographic conditions.				
	Describe the SVP.	☐	☐	
	Explain the effects of Display Bottom Composition on sound propagation.	☐	☐	
	Identify Layer depths.	☐	☐	
	Explain the SVP's effects on sound propagation.	☐	☐	

Score: /112

The trainee must achieve 84/112 SATs AND SATs on all critical items to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 001.07 – Present an Oceanography Brief (B0036)

Presenting an Oceanography Brief

Lesson	Conditions	Sequence
001.07A	Classroom	After 002.01A
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Secret	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Understand implications of Oceanographic information; 2) Describe presentation techniques; and 3) Describe proper speaking practices.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 001.07A lecture on presenting an Oceanography Brief. 2) On completion of the 001.07A lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 001.07A Presentation; 2) Confirmation questions; 3) Classroom; and 4) Example Oceanography Brief.	1) Lesson material.

Notes to Instructor
The student will be assessed on the presentation of an oceanography brief. Focus on presentation techniques and eliminating bad habits.

EO 001.07 – Present an Oceanography Brief (B0036)

Present an Oceanography Brief

Lesson	Conditions	Sequence
001.07B	Classroom/Theatre	001.07A
Time	**Classification**	**Instructor**
3 x 50 Minute Periods	Secret	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Identify oceanography brief components; 2) Use proper presentation techniques; and 3) Draw proper conclusion from oceanographic data.	EDI

Scenario	You are at sea and you are about to reach your submarine patrol area (SPA). The UC1 has handed you the brief created from the information in the environmental briefing folder and asked you to update the oceanography and present the brief to the CO during the evening operations brief. There is intelligence to suggest that there is a Russian Kilo 4B and one Udaloy in the area. On completion of the patrol we are to RV with HMCS Halifax 14 NM east of the SPA. Your oceanographic features analysis (OFA) is dated 18 days ago and the STDA function on BQQ 10 is reduced IOT provide urgent active capability for expedited weapons release.
Activity Steps	1) Students are shown a scenario and an oceanography brief. 2) The instructor will go over the checklist for EC 001.07. 3) Instructor will conduct a discussion on distracting mannerisms and behavior. 4) Students will present a slide to the instructor in preparation for the EC.

Resources	References
1) Oceanography Brief; and 2) Classroom.	1) Lesson Material; and 2) EC 001.07 checklist.

Notes to Instructor
Provide the students with an example of a brief that is not perfect. Ensure there are a few things for the students to see and want to change (yellow writing on a white background for example).

EC 001.07 – Present an Oceanography Brief

Scope	Conditions	Sequence
EO 001.07	Classroom / Theatre	After 001.07B
Time	**Classification**	**Instructor**
1 x 50 Minute Period (per Student)	Secret Can/US	MS-PO1, SONAR Op (Subs)

Scenario	You have been directed to present an oceanography brief on the area the submarine will be operating in in the next phase of the course. You are at sea and you are about to reach your submarine patrol area (SPA). The UC1 has handed you the brief created from the information in the environmental briefing folder and asked you to update the oceanography and present the brief to the CO during the evening operations brief. There is intelligence to suggest that there is a Russian Kilo 4B and one Udaloy in the area. On completion of the patrol we are to RV with HMCS Halifax 14 NM east of the SPA. Your oceanographic features analysis (OFA) is dated 18 days ago and the STDA function on BQQ 10 is reduced IOT provide urgent active capability for expedited weapons release.
Assessment Steps	1) The student is to present an oceanographic brief to command. 2) Student is given the oceanographic conditions, and is required to present that information to command. Student is expected to present the brief in accordance with the practices covered in EO 001.07. Student will have 30 minutes to present the brief to the instructor and the rest of the class. 3) The instructor will not provide direction. Instructor will evaluate as per the EC 001.07 checklist and will provide questions to assess the understanding of the material. 4) Instructor will provide feedback to the individual student, after the brief, in order to review deficiencies and reinforce correct briefing techniques.

Resources	References
1) Classroom / Theatre 2) Miloc Manuel 3) BQQ-10 STDA 4) Computer with Projector	1) Lesson Material 2) NCPM 403 3) IETM

Notes to Instructor

Minimize setup time by building the brief prior to the course start. Students are being assessed on their ability to present an oceanography brief. Creating the brief will be consolidated in the OJPR. Follow and complete the EC 001 checklist: one per student and one per brief. This

information will be used and built upon when performing the duties and responsibilities of a sonar operator on submarines and during the OJPR.

Limitations	No references and no assistance allowed. This knowledge will be consolidated during the OJPR.
Impact of Failure	Remedial Training as required and Rewrite.

Item	EC 001.07 Present an Oceanography Brief	SAT	UNSAT	Remarks
1. Speaking.				
Introduced him/herself.		☐	☐	
Stated the Aim		☐	☐	
Presentation is clear.		☐	☐	
Proper terminology.		☐	☐	
Volume level is audible.		☐	☐	
2. Presentation Technique.				
Acceptable appearance.		☐	☐	
Emphasized key points.		☐	☐	
Maintains good posture.		☐	☐	
Faces the audience.		☐	☐	
Confident and professional bearing.		☐	☐	
Avoided distracting mannerisms/behavior.		☐	☐	
3. Knowledge.				
Has an understanding of the briefed material.		☐	☐	
Material was accurate.		☐	☐	
Concepts were clearly explained.		☐	☐	
Applied principles of oceanographic theory to the material.		☐	☐	
Responded to questions in a professional, non-confrontational manner.		☐	☐	

Score: /16

The trainee must achieve 12/16 SATs to pass this EC.

Positive Feedback		Improvement Feedback	
Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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EO 001.01-06 – Operate Main Frame Sonar

Developmental VCSTT

Lesson	Conditions	Sequence
001.01-06A	VCSTT	After EC 001.04-06 Practical
Time	**Classification**	**Instructor**
28 x 50 Minute Periods	Secret Can/US	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Operate mainframe Sonar in a simulated at sea environment; and 2) Amalgamate and employ previous lesson objectives.	EDI

Scenario	You are off the coast of Halifax, open ocean. You are travelling at 57 meters and expect to return to periscope depth on your watch. Oceanographic conditions are as briefed in EC 001.07. There is moderate merchant traffic, one warship and an un-located threat submarine in the area. Your watch just started and you are turning over the sonar.
Activity Steps	1) The student is to demonstrate an ability operate the main frame sonar by reacting to direction in a simulated at sea environment. 2) Student is given the scenario and access to the VCSTT. Student is expected to set up and conduct equipment checks covered in EO 001.01. The student will cover the operating procedures for active, passive broadband, passive narrowband, intercept, and operate the mainframe sonar logging functions as set out in EOs 001.02-6 while maintaining an awareness of the environmental conditions using the STDA function. 3) The instructor will provide direction as per PC 001 checklist and will ensure the students have a complete grasp of the concepts taught in this PO. 4) This exercise is to be conducted in a simulated watch in order to teach the students how to combine the skills taught in PO 001 and apply them as a watch. 5) The instructor will encourage the students to work together to solve any problems that arise in operating the Sonars. The instructor is permitted to assist, however, teamwork and problem solving between the students is desirable.

Resources	References
1) VCSTT; 2) CFCD 113 (D) AIG; 3) Chronoscope; 4) Fire control Operator; and 5) VCSTT Operator.	1) Lesson Material; 2) IETM; and 3) NCPM 403.

Notes to Instructor

Frame these developmental runs in the form of a watch and have them consist of taking over the watch, operating in the scenario and getting ready to turn over the watch to the next one. Alternate what tasks are given to each student in order to complete all tasks in the PC 001 checklist. This checklist is an aide for the instructor to ensure all points are getting covered in the scenario and not intended as a performance evaluation.

Item	DEVELOPMENTAL VCSTT Operate Main Frame Sonar	SAT	UNSAT	Remarks
1. Perform closing up checks to include.				
Perform a communications check.		☐	☐	
Check cursor response on all sensors.		☐	☐	
Verify audio on all sensors.		☐	☐	
Check the status of arrays.		☐	☐	
Check ship's and array heading follows actual ship's heading.		☐	☐	
Ensure navigation data is correct.		☐	☐	
Verify that the SONSYM is operational.		☐	☐	
Input correct display settings to include screen colour, baffle indicators, ship's heading marker, water falling sierra numbers and sonar logger is operational for following displays: 1. CBB; 2. Multi Hull; 3. PNB; 4. AI&R; and 5. HFA.		☐	☐	
Ensure acoustic information is updating.		☐	☐	
Set broadband vertical beam width to wide.		☐	☐	
Set narrowband normalizers to narrow.		☐	☐	
Check that server temperature levels are within tolerances.		☐	☐	
2. Start STDA and select the following.				
Set submarine position to in situ.		☐	☐	
Switch map display between 2D and 3D.		☐	☐	
Manipulate sound velocity display to include depth and unit of measurements.		☐	☐	
3. Operate HFA.				
Select HFA Mode.		☐	☐	
Select Cycle Test.		☐	☐	
Initiate single transmission in Cycle Test.		☐	☐	
Adjust Range scale.		☐	☐	
Adjust Bearing coverage.		☐	☐	
Switch between sub-modes to include: 1. ASW;		☐	☐	

Item	DEVELOPMENTAL VCSTT Operate Main Frame Sonar	SAT	UNSAT	Remarks
	2. IKA – Depth; 3. IKA – Fade; 4. Mine Avoidance; 5. PD; and 6. PUMA.			
	Switch between display options to include: 1. PPI Intensity; 2. PBB/PPI; 3. B-Scan Single; 4. B-Scan Double; 5. B-Scan Triple; and 6. PPI Depth.	☐	☐	
	Track a contact actively. (ASW/PD)	☐	☐	
	Select Pulse type.	☐	☐	
	Update active contact. (Track Expand Window)	☐	☐	
	Enable/Disable CAD.	☐	☐	
	Set Standoff Range. (Mine Avoidance)	☐	☐	
	Set Safety Corridor. (Mine Avoidance)	☐	☐	
	Operate STDA Thumbnail. (PD/ASW)	☐	☐	
	Correlate Active contact with HFA PBB Wrap Around.	☐	☐	
	Interpret Depth. (IKA/PPI-Depth)	☐	☐	
	Toggle CAD overlays.	☐	☐	
	Toggle External Tracker Repeaters.	☐	☐	
	Switch frequency via Pre-sets. (Processing)	☐	☐	
	Select processing type. (Monopulse/2D Single/2D Triple)	☐	☐	
	Identify Primary, Surface and Bottom Reject contacts. (Monopulse/B-SCSAB)	☐	☐	
	Set Depression Elevation.	☐	☐	
	Use Beam Normalizers to suppress broadband spokes. (PPI)	☐	☐	
	Match Sierra number to Fire Control Number.	☐	☐	
4. Operate passive sub-systems.				
-PBB-				
	Initiate contact:	☐	☐	

Item	DEVELOPMENTAL VCSTT Operate Main Frame Sonar	SAT	UNSAT	Remarks
	1. Auto Tracker Bow/PRS; and 2. Manual Tracker Bow/PRS.			
	Initiate Torpedo drills **(CRITICAL)**	☐	☐	
	Switch between Auto and Manual tracking	☐	☐	
	Determine range from contact and assess tabs.	☐	☐	
	Use track windows to refine PRS solution. (MHD)	☐	☐	
	Calculate Relative Bearing.	☐	☐	
	Switch between PRS FM and MA arrays.	☐	☐	
	Enable/disable Global Overlays.	☐	☐	
	Initiate a Scissor Gram. (CBB)	☐	☐	
	Operate Cursors to include: 1. Crosshair; 2. Delta; and 3. Annotate gram.	☐	☐	
	Switch between multiple arrays. (CBB/MHD)	☐	☐	
	Carry out voice procedures IAW NCPM 403 for: 1. New Contact; 2. Amplify Contact; and 3. Lost Contact.	☐	☐	
	Carry out Close Quarters Drill. **(CRITICAL)**	☐	☐	
	Switch between frequency bandwidths. (CBB/MHD)	☐	☐	
	Locate and relay Non Acoustic Data to include: 1. Date/Time; 2. Heading; 3. Depth; 4. Speed; 5. Pitch/Roll; 6. Keel to Bottom Depth; 7. Sound Velocity; and 8. Array Heading (if fitted).	☐	☐	
	Toggle specific overlays. (CBB/MHD)	☐	☐	
	Toggle contact symbols.	☐	☐	

Item	DEVELOPMENTAL VCSTT Operate Main Frame Sonar	SAT	UNSAT	Remarks
Interrogate Feature CF.		☐	☐	
Recall and store display configurations.		☐	☐	
Set Hybrid Contact Follower center frequency and bandwidth.		☐	☐	
Manipulate Audio Filters.		☐	☐	
Set Active Rejection band filter center frequency and bandwidth.		☐	☐	
Rig for black.		☐	☐	
Access calculator.		☐	☐	
Measure bearing.		☐	☐	
Measure Bearing Rate. (Bow/PRS)		☐	☐	
Determine Intensity.		☐	☐	
Edit contact history via CF stitching functions.		☐	☐	
Interrogate acoustic energy to identify contacts.		☐	☐	
Conduct Stern arc clearance (MHD)		☐	☐	
Select and deselect Grams displayed. (CBB)		☐	☐	
Adjust Update Rate. (CBB/MHD)		☐	☐	
Switch Vertical Beam Width. (CBB/MHD)		☐	☐	
Conduct Triangulation Ranging.		☐	☐	
Drill Down frequency bandwidths. (CBB)		☐	☐	
Recognize target Zig.		☐	☐	
-PNB-				
Track a NB Contact via Vertical Grams.		☐	☐	
Switch formats between: 1. Full Frequency; 2. Paged; 3. Demon; 4. Linked LOFAR Gram; 5. Scissor Gram; and 6. Fastgram.		☐	☐	
Set input source. (Fastgram)		☐	☐	
Set frequency BW and center frequency. (Fastgram)		☐	☐	
Select and center ALI display. (Fastgram)		☐	☐	

Item	DEVELOPMENTAL VCSTT Operate Main Frame Sonar	SAT	UNSAT	Remarks
Determine B/S configuration and Rev count.		☐	☐	
Send Classification information to Log.		☐	☐	
Switch between contacts on Linked Lofar Gram.		☐	☐	
Switch input source between multiple hydrophone arrays.		☐	☐	
Manipulate cursor style and colour.		☐	☐	
Identify Hybrid Contact Follower bandwidth and center frequency.		☐	☐	
Interpret cursor read our area to include: 1. Frequency; 2. Bearing; 3. Time; 4. Δ Time of event; and 5. Δ Frequency.		☐	☐	
Use cursers to include: 1. Crosshair; 2. Harmonic; 3. Comb; and 4. Delta.		☐	☐	
Change update rate.		☐	☐	
Verify Normalizers are set to narrow.		☐	☐	
Select Demon bandwidth.		☐	☐	
Update Vertical Gram position.		☐	☐	
Scroll through narrowband bearings with: 1. Paged; 2. Full Frequency Single; 3. Demon; 4. 8 Ored; and 5. 16 Ored.		☐	☐	
Manipulate the Bearing tape to reposition Analysis Grams and Vertical Grams.		☐	☐	
Correlate BB and NB contacts.		☐	☐	
Change Vertical Beamwidth.		☐	☐	
Initiate Torpedo drills **(CRITICAL)**		☐	☐	
Drop Contact.		☐	☐	
Regain Contact.		☐	☐	

Item	DEVELOPMENTAL VCSTT Operate Main Frame Sonar	SAT	UNSAT	Remarks
Recognize NB indication of target Zig.		☐	☐	
Identify bearing movement based on coloured overlays.		☐	☐	
-AI&R-				
Select AI&R display.		☐	☐	
Switch between display formats.		☐	☐	
Switch between true and relative bearing.		☐	☐	
Operate cursors to include: 1. Scroll; 2. Crosshair; 3. Delta; and 4. Zoom.		☐	☐	
Switch between input sources.		☐	☐	
Set and select Auxiliary inputs.		☐	☐	
Select audio source hydrophones.		☐	☐	
Recognize hydrophone status via indicators.		☐	☐	
Relay cursor readout information.		☐	☐	
Cycle quick lineup buttons to include: 1. Normal; 2. PD; and 3. Torp.		☐	☐	
Switch between Bearing/Frequency displays.		☐	☐	
Change update rates.		☐	☐	
Select track controls to include: 1. BTH; 2. SPLTH; 3. FBI; and 4. FTH.		☐	☐	
Measure pulse length.		☐	☐	
Conduct Ping Stolen Range.		☐	☐	
Adjust Frequency bandwidth and center frequency.		☐	☐	
Switch between Contact and Event history.		☐	☐	
Promote active emissions to a Papa tracker.		☐	☐	
Promote active emissions to a Sierra tracker.		☐	☐	

Item	DEVELOPMENTAL VCSTT Operate Main Frame Sonar	SAT	UNSAT	Remarks
Clear History.		☐	☐	
Set an Alert Band.		☐	☐	
Configure contact association for bearing and frequency.		☐	☐	
Set detector sensitivity.		☐	☐	
Set Own Ship Muting on/off.		☐	☐	
Configure Scanner.		☐	☐	
Set Quick Line Up to WL/UC1 settings.		☐	☐	
5. Operate noise monitoring equipment.				
Select noise monitoring.		☐	☐	
Select hydrophone/array.		☐	☐	
Manipulate frequency bandwidth and center frequency.		☐	☐	
Select Self noise monitoring.		☐	☐	
Cycle through sensors.		☐	☐	
6. Record acoustic intelligence data.				
Open the DDCS window.		☐	☐	
Configure Hydrophone Inputs.		☐	☐	
Configure Sail recorder input.		☐	☐	
Assign a tape to: 1. DDCS Hyd; 2. AEMP; and 3. Tools.		☐	☐	
Open Recording Tab.		☐	☐	
Initiate event recording/continuous recording.		☐	☐	
Copy a tape via Media utilities.		☐	☐	
Take a Screen capture via Menubar and Alert Bar.		☐	☐	
Recall the screen capture via the Screen Capture Viewer.		☐	☐	
Manipulate the Zoom function in the Screen Capture Viewer.		☐	☐	
Burn Screen Capture to disk.		☐	☐	
Annotate an event with the Sonar Logger.		☐	☐	
Annotate a contact with the Sonar Logger.		☐	☐	

Item	DEVELOPMENTAL VCSTT Operate Main Frame Sonar	SAT	UNSAT	Remarks
Start the Scribe recorder.		☐	☐	
Lease an LTO4 tape to the Scribe Recorder and record HFA.		☐	☐	
7. Assess oceanographic conditions.				
Describe the SVP.		☐	☐	
Explain the effects of Display Bottom Composition on sound propagation.		☐	☐	
Identify Layer depths.		☐	☐	
Explain the SVP's effects on sound propagation.		☐	☐	

Score: / 151

The trainee must achieve 114/151 SATs AND SATs on all critical items to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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PC 001A – Operate Ancillary Sonars

Scope	Conditions	Sequence
PO 001	Classroom	EC 001.07
Time	**Classification**	**Instructor**
4 x 50 Minute Periods	Secret Can/US	MS-PO1, SONAR Op (Subs)

Scenario	You are off the coast of Halifax, open ocean. You are travelling at 57 meters and expect to return to periscope depth on your watch. There is moderate merchant traffic and one warship in the area. You have been tasked to operate the AN/BQQ-10 Sonar.
Assessment Steps	1) The student is to demonstrate a knowledge of the capabilities and limitations of the ANQ BQQ 10 sonar as taught in PO 001 by answering the knowledge based questions correctly. 2) Student is given the scenario and questionnaire. Student is expected to answer the knowledge based questions correctly. The student has 40 minutes to complete the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) PC 001 Questionnaire; 2) PC 001 Answer Guide; 3) PP&S; and 4) Classroom.	1) Lesson Material; 2) NCPM 403; and 3) IETM.

Notes to Instructor
This material is to be used every time the students operate the mainframe sonar in the VCSST and subsequently on board.

Limitations	No references or assistance will be provided during the PC.
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

PC 001B – Operate Mainframe SONAR

Scope	Conditions	Sequence
PO 001	VCSTT	After EC 001.04-06 Practical
Time	**Classification**	**Instructor**
4 x 50 Minute Periods (per Student)	Secret Can/US	MS-PO1 SONAR Op (Subs)

Scenario	You are off the coast of Halifax, open ocean. You are travelling at 57 meters and expect to return to periscope depth on your watch. Oceanographic conditions are as briefed in EC 001.07. There is moderate merchant traffic and one warship in the area.
Assessment Steps	1) The student is to demonstrate an ability operate the main frame sonar by reacting to direction in a simulated at sea environment. 2) Student is given the scenario and access to the VCSTT. Student is expected to set up and conduct equipment checks covered in EO 001.01. The student cover the operating procedures for active, passive broadband, passive narrowband, intercept, and operate the mainframe sonar logging functions while maintaining an awareness of the environmental conditions using the STDA function. 3) The instructor will provide direction as per PC 001 checklist and may clarify any directions that are not fully understood. 4) Instructor will provide feedback to the individual student, after each scenario, in order to review the tasks and reinforce correct operating procedures.

Resources	References
1) VCSTT; 2) CFCD 113 (E) AIG; 3) Chronoscope; 4) Fire control Operator; and 5) VCSTT Operator.	1) Lesson Material; 2) NCPM 403; and 3) IETM.

Notes to Instructor

Minimize setup time by building scenarios before commencing the trainer phase to include counter-detection and torpedoes firing. Verify with the trainer staff that the trainer time is indeed booked. Follow and fill the PC 001 checklist: one per student and one per run. This information will be used and built upon when performing the duties and responsibilities of a sonar operator on submarines and during the OJPR.

Limitations	No assistance will be provided during the PC.
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

Item	PC 001B Operate Main Frame Sonar	SAT	UNSAT	Remarks
1. Perform closing up checks to include.				
Perform a communications check.		☐	☐	
Check cursor response on all sensors.		☐	☐	
Verify audio on all sensors.		☐	☐	
Check the status of arrays.		☐	☐	
Check ship's and array heading follows actual ship's heading.		☐	☐	
Ensure navigation data is correct.		☐	☐	
Verify that the SONSYM is operational.		☐	☐	
Input correct display settings to include screen colour, baffle indicators, ship's heading marker, water falling sierra numbers and sonar logger is operational for following displays: 1. CBB; 2. Multi Hull; 3. PNB; 4. AI&R; and 5. HFA.		☐	☐	
Ensure acoustic information is updating.		☐	☐	
Set broadband vertical beam width to wide.		☐	☐	
Set narrowband normalizers to narrow.		☐	☐	
Check that server temperature levels are within tolerances.		☐	☐	
2. Start STDA and select the following.				
Set submarine position to in situ.		☐	☐	
Switch map display between 2D and 3D.		☐	☐	
Manipulate sound velocity display to include depth and unit of measurements.		☐	☐	
3. Operate HFA.				
Select HFA Mode.		☐	☐	
Select Cycle Test.		☐	☐	

Item	PC 001B Operate Main Frame Sonar	SAT	UNSAT	Remarks
Initiate single transmission in Cycle Test.		☐	☐	
Adjust Range scale.		☐	☐	
Adjust Bearing coverage.		☐	☐	
Switch between sub-modes to include: 1. ASW; 2. IKA – Depth; 3. IKA – Fade; 4. Mine Avoidance; 5. PD; and 6. PUMA.		☐	☐	
Switch between display options to include: 1. PPI Intensity; 2. PBB/PPI; 3. B-Scan Single; 4. B-Scan Double; 5. B-Scan Triple; and 6. PPI Depth.		☐	☐	
Track a contact actively. (ASW/PD)		☐	☐	
Select Pulse type.		☐	☐	
Update active contact. (Track Expand Window)		☐	☐	
Enable/Disable CAD.		☐	☐	
Set Standoff Range. (Mine Avoidance)		☐	☐	
Set Safety Corridor. (Mine Avoidance)		☐	☐	
Operate STDA Thumbnail. (PD/ASW)		☐	☐	
Correlate Active contact with HFA PBB Wrap Around.		☐	☐	
Interpret Depth. (IKA/PPI-Depth)		☐	☐	
Toggle CAD overlays.		☐	☐	
Toggle External Tracker Repeaters.		☐	☐	
Switch frequency via Pre-sets. (Processing)		☐	☐	
Select processing type. (Monopulse/2D Single/2D Triple)		☐	☐	
Identify Primary, Surface and Bottom Reject contacts. (Monopulse/B-SCSAB)		☐	☐	
Set Depression Elevation.		☐	☐	
Use Beam Normalizers to suppress broadband spokes. (PPI)		☐	☐	

Item	PC 001B Operate Main Frame Sonar	SAT	UNSAT	Remarks
Match Sierra number to Fire Control Number.		☐	☐	
4. Operate passive sub-systems.				
-PBB-				
Initiate contact: 1. Auto Tracker Bow/PRS; and 2. Manual Tracker Bow/PRS.		☐	☐	
Initiate Torpedo drills **(CRITICAL)**		☐	☐	
Switch between Auto and Manual tracking		☐	☐	
Determine range from contact and assess tabs.		☐	☐	
Use track windows to refine PRS solution. (MHD)		☐	☐	
Calculate Relative Bearing.		☐	☐	
Switch between PRS FM and MA arrays.		☐	☐	
Enable/disable Global Overlays.		☐	☐	
Initiate a Scissor Gram. (CBB)		☐	☐	
Operate Cursors to include: 1. Crosshair; 2. Delta; and 3. Annotate gram.		☐	☐	
Switch between multiple arrays. (CBB/MHD)		☐	☐	
Carry out voice procedures IAW NCPM 403 for: 1. New Contact; 2. Amplify Contact; and 3. Lost Contact.		☐	☐	
Carry out Close Quarters Drill. **(CRITICAL)**		☐	☐	
Switch between frequency bandwidths. (CBB/MHD)		☐	☐	
Locate and relay Non Acoustic Data to include: 1. Date/Time; 2. Heading; 3. Depth; 4. Speed; 5. Pitch/Roll;		☐	☐	

Item	PC 001B Operate Main Frame Sonar	SAT	UNSAT	Remarks
	6. Keel to Bottom Depth; 7. Sound Velocity; and 8. Array Heading (if fitted).			
	Toggle specific overlays. (CBB/MHD)	☐	☐	
	Toggle contact symbols.	☐	☐	
	Interrogate Feature CF.	☐	☐	
	Recall and store display configurations.	☐	☐	
	Set Hybrid Contact Follower center frequency and bandwidth.	☐	☐	
	Manipulate Audio Filters.	☐	☐	
	Set Active Rejection band filter center frequency and bandwidth.	☐	☐	
	Rig for black.	☐	☐	
	Access calculator.	☐	☐	
	Measure bearing.	☐	☐	
	Measure Bearing Rate. (Bow/PRS)	☐	☐	
	Determine Intensity.	☐	☐	
	Edit contact history via CF stitching functions.	☐	☐	
	Interrogate acoustic energy to identify contacts.	☐	☐	
	Conduct Stern arc clearance (MHD)	☐	☐	
	Select and deselect Grams displayed. (CBB)	☐	☐	
	Adjust Update Rate. (CBB/MHD)	☐	☐	
	Switch Vertical Beam Width. (CBB/MHD)	☐	☐	
	Conduct Triangulation Ranging.	☐	☐	
	Drill Down frequency bandwidths. (CBB)	☐	☐	
	Recognize target Zig.	☐	☐	
-PNB-				
	Track a NB Contact via Vertical Grams.	☐	☐	
	Switch formats between: 1. Full Frequency; 2. Paged; 3. Demon; 4. Linked LOFAR Gram; 5. Scissor Gram; and	☐	☐	

Item	PC 001B Operate Main Frame Sonar	SAT	UNSAT	Remarks
6. Fastgram.				
Set input source. (Fastgram)		☐	☐	
Set frequency BW and center frequency. (Fastgram)		☐	☐	
Select and center ALI display. (Fastgram)		☐	☐	
Determine B/S configuration and Rev count.		☐	☐	
Send Classification information to Log.		☐	☐	
Switch between contacts on Linked Lofar Gram.		☐	☐	
Switch input source between multiple hydrophone arrays.		☐	☐	
Manipulate cursor style and colour.		☐	☐	
Identify Hybrid Contact Follower bandwidth and center frequency.		☐	☐	
Interpret cursor read our area to include: 1. Frequency; 2. Bearing; 3. Time; 4. Δ Time of event; and 5. Δ Frequency.		☐	☐	
Use cursers to include: 1. Crosshair; 2. Harmonic; 3. Comb; and 4. Delta.		☐	☐	
Change update rate.		☐	☐	
Verify Normalizers are set to narrow.		☐	☐	
Select Demon bandwidth.		☐	☐	
Update Vertical Gram position.		☐	☐	
Scroll through narrowband bearings with: 1. Paged; 2. Full Frequency Single; 3. Demon; 4. 8 Ored; and 5. 16 Ored.		☐	☐	
Manipulate the Bearing tape to reposition Analysis Grams and Vertical Grams.		☐	☐	
Correlate BB and NB contacts.		☐	☐	

Item	PC 001B Operate Main Frame Sonar	SAT	UNSAT	Remarks
Change Vertical Beamwidth.		☐	☐	
Initiate Torpedo drills **(CRITICAL)**		☐	☐	
Drop Contact.		☐	☐	
Regain Contact.		☐	☐	
Recognize NB indication of target Zig.		☐	☐	
Identify bearing movement based on coloured overlays.		☐	☐	
-AI&R-				
Select AI&R display.		☐	☐	
Switch between display formats.		☐	☐	
Switch between true and relative bearing.		☐	☐	
Operate cursors to include: 1. Scroll; 2. Crosshair; 3. Delta; and 4. Zoom.		☐	☐	
Switch between input sources.		☐	☐	
Set and select Auxiliary inputs.		☐	☐	
Select audio source hydrophones.		☐	☐	
Recognize hydrophone status via indicators.		☐	☐	
Relay cursor readout information.		☐	☐	
Cycle quick lineup buttons to include: 1. Normal; 2. PD; and 3. Torp.		☐	☐	
Switch between Bearing/Frequency displays.		☐	☐	
Change update rates.		☐	☐	
Select track controls to include: 1. BTH; 2. SPLTH; 3. FBI; and 4. FTH.		☐	☐	
Measure pulse length.		☐	☐	
Conduct Ping Stolen Range.		☐	☐	

Item	PC 001B Operate Main Frame Sonar	SAT	UNSAT	Remarks
Adjust Frequency bandwidth and center frequency.		☐	☐	
Switch between Contact and Event history.		☐	☐	
Promote active emissions to a Papa tracker.		☐	☐	
Promote active emissions to a Sierra tracker.		☐	☐	
Clear History.		☐	☐	
Set an Alert Band.		☐	☐	
Configure contact association for bearing and frequency.		☐	☐	
Set detector sensitivity.		☐	☐	
Set Own Ship Muting on/off.		☐	☐	
Configure Scanner.		☐	☐	
Set Quick Line Up to WL/UC1 settings.		☐	☐	
5. Operate noise monitoring equipment.				
Select noise monitoring.		☐	☐	
Select hydrophone/array.		☐	☐	
Manipulate frequency bandwidth and center frequency.		☐	☐	
Select Self noise monitoring.		☐	☐	
Cycle through sensors.		☐	☐	
6. Record acoustic intelligence data.				
Open the DDCS window.		☐	☐	
Configure Hydrophone Inputs.		☐	☐	
Configure Sail recorder input.		☐	☐	
Assign a tape to: 1. DDCS Hyd; 2. AEMP; and 3. Tools.		☐	☐	
Open Recording Tab.		☐	☐	
Initiate event recording/continuous recording.		☐	☐	
Copy a tape via Media utilities.		☐	☐	
Take a Screen capture via Menubar and Alert Bar.		☐	☐	
Recall the screen capture via the Screen Capture Viewer.		☐	☐	

Item	PC 001B Operate Main Frame Sonar	SAT	UNSAT	Remarks
Manipulate the Zoom function in the Screen Capture Viewer.		☐	☐	
Burn Screen Capture to disk.		☐	☐	
Annotate an event with the Sonar Logger.		☐	☐	
Annotate a contact with the Sonar Logger.		☐	☐	
Start the Scribe recorder.		☐	☐	
Lease an LTO4 tape to the Scribe Recorder and record HFA.		☐	☐	
7. Assess oceanographic conditions.				
Describe the SVP.		☐	☐	
Explain the effects of Display Bottom Composition on sound propagation.		☐	☐	
Identify Layer depths.		☐	☐	
Explain the SVP's effects on sound propagation.		☐	☐	

Score: / 151

The trainee must achieve 114/151 SATs AND SATs on all critical items to pass this PC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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PO 002 – Operate ancillary acoustic equipment

1. **Performance**: Operate ancillary acoustic equipment (C0025)

2. **Conditions**:

   a. Given:

      (1) Chronoscope;
      (2) Supervision;
      (3) Logs;
      (4) Expendable stores;
      (5) Sonars; and
      (6) References.

   b. Denied: Nil.

   c. Environment:

      (1) At sea or alongside;
      (2) Night or day; and
      (3) Any weather conditions.

3. **Standard**: The Submarine Acoustic Operator shall operate ancillary acoustic equipment to include:

   a. Operating underwater comms (T0012) to include:

      (1) Operating the 2008 Underwater telephone (UWT) by:
         i. Performing Uncle Joe procedures/Authenticating communication transmissions (A0030);
         ii. Performing Mark Snap procedures;
         iii. Performing long post diving checks;
         iv. Returning to periscope depth;
         v. Conducting proximity checks;
         vi. Performing a bearing investigation;
         vii. Close quarters drills; and
         viii. Detect identify and track contacts.

      (2) Operating the 2073 E-UWT to include:
         i. Operating during an escape;
         ii. Operating during long post diving checks;
         iii. Operating via external battery; and
         iv. Performing system flash up and shut down.

   b. Operating echo sounders (T0013); and

   c. Briefing the OOW/CO on XBT/XSV Forms by:

(1) Operating the MK 8F to include:
   i. Verifying underwater expendable inventories (C0055);
   ii. Preparing underwater expendables for launch (C0050);
   iii. Launching underwater expendables (C0035).
   iv. Obtain XBT/XSV forms; and,
   v. Analyzing XBT/XSV forms.

(2) Operating velocimeter (T0014) by:
   i. Performing flash up and shut down;
   ii. Obtaining SV profiles; and
   iii. Analyzing SV profiles.

4. **Training Limitations**: Nil.

EO 002.01 - Operate Underwater Communications (T0012)

Introduction to the 2008 Underwater Telephone (UWT)

Lesson	Conditions	Sequence
002.01A	Classroom	After EC 001.04-6
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Confidential	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Identify purpose of 2008; 2) Describe capabilities and limitations of 2008; 3) Describe physical components; 4) Describe operating procedure of 2008; 5) Explain types of voice report; and 6) Explain voice report procedure.	Interactive Lecture

Scenario	Nil
Activity Steps	1) Instructor will present the 002.01A lecture on 2008 UWT operations, capabilities and limitations. 2) On completion of the 2008 UWT lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 002.01A Presentation; 2) Confirmation questions; and 3) Classroom.	1) Lesson Material; 2) NCPM 403; 3) AXP 1 (E); 4) C-69-846-000/MS-001 2008 Manual; and 5) BR 8070.

Notes to Instructor
This portion of PO 002 is taught during PO 001 in order to prepare students to operate the 2008 during the trainer phase of the PO 001.

EO 002.01 - Operate Underwater Communications (T0012)

Introduction to the 2073 E-UWT

Lesson	Conditions	Sequence
002.01B	Classroom	After PC 001B
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Confidential	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Describe purpose of 2073; 2) Explain functions of 2073; 3) Identify capabilities and limitations of 2073; and 4) Describe operating procedure of 2073.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 002.01B lecture on the 2073 EUWT operations, capabilities and limitations. 2) On completion of the 2073 lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 002.01B Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) CFCD 102; 3) NCPM 403; 4) C-69-860-000/MS-001 2073 Handbook; and 5) BR 8070.

Notes to Instructor

Ensure that the students understand that they, as Sonar Ops, are required to be the SME with regards to this equipment. In an escape situation, any Sonar Op may be the unsupervised operator of this equipment.

EO 002.01 - Operate Underwater Communications (T0012)

Operate the 2008 UWT

Lesson	Conditions	Sequence
002.01C	VCSTT	After 001.07B
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Confidential	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Operate 2008 UWT; 2) Perform Uncle Joe procedures; 3) Perform Mark Snap procedures; and 4) Perform Long post diving checks.	EDI

Scenario	This EDI will be conducted during the PO 001 Trainer Phase and will be take place during various scenarios. As at sea, the BQQ-10 8001 operator is to close up IAW NCPM 403.
Activity Steps	1) Instructor will demonstrate the correct operating procedure for all evolutions listed in NCPM 403. 2) Instructor will have each student close up the 2008 for all evolutions dictated in NCPM 403. 3) Throughout the scenarios conducted in the PO 001 trainer phase, each student will be required to close up this position and perform all operating procedures. 4) Student will carry out all operations on the 2008 as listed in EC 002.01 checklist.

Resources	References
1) VCSTT; and 2) Operational 2008 UWT.	1) Lesson Material; 2) NCPM 403; 3) AXP1 (E); 4) C-69-846-000/MS-001 Manual; and 5) BR 8070.

Notes to Instructor
The lecture for this EO is taught during PO 001 in preparation for the VCSTT phase of PO 001.

EO 002.01 - Operate Underwater Communications (T0012)

Operate the 2073 E-UWT

Lesson	Conditions	Sequence
002.01 D	TWS	After 002.03B
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Confidential	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) System familiarization of 2073; 2) Perform system flash up; 3) Operate 2073 in an escape scenario; 4) Operate 2073 during long post diving checks; 5) Operate 2073 via external battery; and 6) Perform system shut down.	EDI

Scenario	Nil.
Activity Steps	1) Instructor will introduce the students to the 2073 EUWT. 2) Instructor will demonstrate flash up, long post diving checks, modes of operation, battery change over and system shut down. 3) Students will simulate flash up, long post diving checks, transmit in all modes, connect the external battery and perform system shut down.

Resources	References
1) TWS; and 2) Operational 2073 E-UWT.	1) Lesson Material; 2) CFCD 102; 3) NCPM 403; 4) C-69-860-000/MS-001, 2073 Handbook; and 5) BR 8070.

Notes to Instructor
Ensure that the students understand that they, as Sonar Ops, are required to be the SME with regards to this equipment. In an escape situation, any Sonar Op may be the unsupervised operator of this equipment.

EC 002.01 – Operate Underwater Communications (T0012)

Scope	Conditions	Sequence
EO 002.01	TWS	After trainer phase EDIs
Time	**Classification**	**Instructor**
1 x 50 Minute Period (per Student)	Confidential	MS-PO1 SONAR Op (Subs)

Scenario	You are off the coast of Halifax, open ocean. You are travelling at 57 meters and expect to return to periscope depth on your watch. There is moderate merchant traffic and one warship in the area. You have been tasked as the 2008 operator and an abrupt in contact collides with the submarine during the RTPD. The submarine sinks in 90m of water and the pressure hull is intact. You are in the after escape compartment and are tasked by the senior survivor to operate the 2073. There is a DSRV and warship on station engaged in a rescue operation.
Assessment Steps	1) The student is to demonstrate an ability set up the underwater telephones and operate them in accordance with NCPM 403 and CFCD 103. By reacting to events in a simulated at sea environment. 2) Student is given the scenario and access to the TWS. Student is expected to set up and conduct equipment checks covered in EO 002.01. The student will cover the operating procedures for the 2008 and the 2073.Students will have 40 minutes to complete the exam. 3) The instructor will provide direction as per EC 002.01 checklist and may clarify any directions that are not fully understood. 4) Instructor will provide feedback to the individual student, after each scenario, in order to review the tasks and reinforce correct operating procedures.

Resources	References
1) TWS; 2) CFCD 113 (E) AIG; 3) Chronoscope; 4) Guard Cards; and 5) AXP 1 (E).	1) Lesson Material; 2) NCPM 403; and 3) Guard Cards.

Notes to Instructor
Verify with the TWS staff that the trainer time is indeed booked. Follow and fill the EC 002.01 checklist: one per student and one per run. This information will be used and built upon when performing the duties and responsibilities of a sonar operator on submarines and during the OJPR.

Limitations	No assistance will be provided during the EC. Guard Cards are to be available to the student.
Impact of Failure	Remedial Training as required and Rewrite.

Item	EC 002.1 Operate Underwater Telephones	SAT	UNSAT	Remarks
1. Operate the 2008 to include.				
Carry out flash up.		☐	☐	
Carry out short and long post diving checks to include: 1. Lighting and power on lights illuminate; 2. Trainable transducer responds to hand wheel; 3. Submarine's heading matches actual heading; 4. Ensure hydrophone audio is correct; 5. Ensure loudspeaker is operational and set to CO's settings; 6. Chronoscope, log and tac-pac are available; and 7. Establish comms with the fwd/aft 2073 using the correct transducers.		☐	☐	
Identify a contact.		☐	☐	
Describe 2008 components and their locations.		☐	☐	
Describe the 2008 controls and functions.		☐	☐	
Conduct a mark snap procedure.		☐	☐	
Conduct Uncle Joe procedure.		☐	☐	
Input standard settings for passive and active operations, to include: 1. TX power; 2. TD selector switch;		☐	☐	

Item	EC 002.1 Operate Underwater Telephones	SAT	UNSAT	Remarks
3. Mode selection switch; 4. Gain mode; 5. Doppler correction; 6. Headset volume; 7. Gain control; 8. Loudspeaker volume; 9. Trainable transducer selector; 10. Auto track; and 11. TCU Power.				
Demonstrate correct voice procedures to include: 1. Closing up report; 2. Report a new contact; 3. Perform an amplifying report; 4. Conduct a lost contact report; 5. Bearing investigation; And 6. Torpedo detection.		☐	☐	
Carry out shut down.		☐	☐	
2. Operate the 2073 EUWT.				
Conduct flash up.		☐	☐	
Carry out short and long post diving checks to include: 1. Verify channel 10 voice is selected; 2. Establish comms with the 2008; 3. Verify hydrophones correct aurally; And 4. Verify channel selection and mode selection lights illuminate when pushed.		☐	☐	
Transmit HF ping.		☐	☐	
Transmit LF ping.		☐	☐	
Transmit SST.		☐	☐	
Transmit HF comms (DSRV) Channel 27		☐	☐	
Transmit HF comms (Divers) Channel 43		☐	☐	
Operate 2073 via external battery.		☐	☐	

Item	EC 002.1 Operate Underwater Telephones	SAT	UNSAT	Remarks

Score: /18

The trainee must achieve 13/18 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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EO 002.02 - Operate Echo Sounders (T0013)

Introduction to the 778/780 Echo Sounder

Lesson	Conditions	Sequence
002.02 A	Classroom	After EO 002.01B
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Confidential	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Explain purpose of the echo sounders; 2) Identify capabilities and limitations of the 778; 3) Identify capabilities and limitations of the 780; 4) Describe the physical components of the 778; 5) Describe the physical components of the 780; 6) Describe the functions of the 778; 7) Describe the functions of the 780; 8) Describe operating procedures of the 778; 9) Describe operating procedures of the 780; 10) Describe the system checks for the 778; and 11) Describe the system checks for the 780.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 002.02A lecture on 778/780 operations, capabilities and limitations. 2) On completion of the 778/780 lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 002.02 A Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) NCPM 403; 3) C-57-803-000/MS-001, 778 Manual; 4) C-57-809-000/ms-001 780 manual; and 5) BR 8070.

Notes to Instructor
Due to equipment availability ashore to the 778/780 Echo Sounders operation be consolidated during the OJPR.

EC 002.02 – Operate Echo Sounders

Scope	Conditions	Sequence
EO 002.02	Classroom	After 002.02A
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Confidential	MS-PO1, SONAR Op (Subs)

Scenario	You are sailing in the position of a Sound Room Watch keeper. The watch leader directs you to operate the 778/780 Echo Sounders. You are to operate both Echo Sounders in accordance with NCPM 403.
Assessment Steps	1) The student is to demonstrate the knowledge of the capabilities and limitations of the 778/780 Echo Sounders taught in EO 002.02 by answering the knowledge based questions correctly. 2) Student is given the scenario and questionnaire. Student is expected to answer the knowledge based questions correctly. The student has 30 minutes to complete the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) EC 002.02 Questionnaire; 2) EC 002.02 Answer key; 3) PP&S; and 4) Classroom.	1) Lesson Material; and 2) NCPM 403.

Notes to Instructor
This material is to be used every time the students conduct training in the VCSTT and subsequently on board.

Limitations	No references and no assistance allowed. This knowledge will be consolidated during the OJPR.
Impact of Failure	Remedial Training as required and Rewrite.

EO 002.03 - Brief the OOW/CO on XBT/XSV Forms

Introduction to the 2004 Velocimeter (T0014)

Lesson	Conditions	Sequence
002.03 A	Classroom	After EO 002.02 A
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Confidential	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Explain purpose of the 2004; 2) Identify capabilities and limitations of the 2004; 3) Describe physical components of the 2004; 4) Describe manning for the 2004; and 5) Explain operating procedures of the 2004.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 002.03A lecture on 2004 operations, components, capabilities and limitations. 2) On completion of the 2004 lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 002.03A Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) NCPM 403; 3) C-69-852-000\MS-001, 2004 Handbook; and 4) C-69-852-000\MS-Z01.

Notes to Instructor
Make sure students have a firm understanding on how to read the 2004 bathy cards while in use. As well as how the information given effects the submarine's buoyancy and sensors.

EO 002.03 - Brief the OOW/CO on XBT/XSV Forms

Analyse XBT/XSV Forms (C0040) and MK 8F

Lesson	Conditions	Sequence
002.03 B	Classroom	After EO 002.03 A
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Unclassified	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Explain purpose of the MK 8F; 2) Identify capabilities and limitations of the MK 8F; 3) Describe physical components of the MK 8F; 4) Identify type of underwater expendables; 5) Describe functions of MK 8F; and 6) Describe operating procedure of MK 8F.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 002.03B lecture on MK 8F operations, capabilities and limitations. 2) On completion of the MK 8F lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 002.03B Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) NCPM 403; and 3) Lockheed martin submarine recorder training.

Notes to Instructor
Due to in ability to link a SSE to the MK 8F in the TWS. The complete operation of this equipment will be consolidated during the OJPR.

EO 002.03 - Brief the OOW/CO on XBT/XSV Forms

Operate Velocimeter (T0014)

Lesson	Conditions	Sequence
002.03C	TWS	After 003.03B
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Unclassified	MS-PO2 Sonar Op (Subs)

Learning Objectives	Method
1) Change the card on the 2004; and 2) Familiarization of controls on the 2004.	EDI

Scenario	Nil.
Activity Steps	1) Instructor will present the 2004 to the students so they can see what the equipment looks like. 2) Instructor will demonstrate how to turn on the equipment, input desired settings and how to change the bathy card. 3) Students will each simulate flash up, set up and changing the card.

Resources	References
1) TWS; and 2) 2004.	1) Lesson material; 2) NCPM 403; and 3) C-69-852-000\MS-001, 2004 Handbook.

Notes to Instructor
The system in the TWS does not have a sound velocity feed, it can be used for changing the paper only. Complete system operation will be conducted during the OJPR.

EC 002.03 – Operate Ancillary Oceanographic Equipment

Scope	Conditions	Sequence
EO 002.03	TWS	After 002.03B
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Confidential	MS-PO1 SONAR Op (Subs)

Scenario	You are off the coast of Halifax, open ocean. You are travelling at 57 meters and expect to return to periscope depth on your watch. You have been tasked as the 2004 operator for the return to periscope depth. On completion of the RTPD you are tasked to launch a SSXBT.
Assessment Steps	1) The student is to demonstrate knowledge of the capabilities and limitations of the 2004 and MK8F 2) Student is given the scenario and access to the TWS. Student is expected to answer questions pertaining to the capabilities and limitations of the 2004 and the MK 8F. Student has 30 minutes to write the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) EC 002.03 Questionnaire; 2) EC 002.03 Answer Guide; 3) PP&S; and 4) Classroom.	1) Lesson Material; 2) NCPM 403; and 3) VCSOPs.

Notes to Instructor
Verify with the TWS staff that the trainer time is booked. Follow and fill the EC 002.03 checklist: one per student and one per run. This information will be used and built upon when performing the duties and responsibilities of a sonar operator on submarines and during the OJPR.

Limitations	No assistance will be provided during the EC.
Impact of Failure	Remedial Training as required and Rewrite.

PC 002 – Operate Ancillary Acoustic Equipment

Scope	Conditions	Sequence
PO 002	Classroom	After EC 002.03
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Confidential	MS-PO1 SONAR Op (Subs)

Scenario	You are sailing in the position of a Sound Room Watch keeper. The watch leader directs you to operate various ancillary sonars in accordance with NCPM 403.
Assessment Steps	1) The student is to demonstrate a knowledge of the capabilities and limitations of the ancillary sonars as taught in PO 002 by answering the knowledge based questions correctly. 2) Student is given the scenario and questionnaire. Student is expected to answer the knowledge based questions correctly. The student has 40 minutes to complete the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) EC 002.01 questionnaire; 2) EC 002.02 questionnaire; 3) PP&S; and 4) Classroom.	1) Lesson Material; and 2) NCPM 403.

Notes to Instructor
This material is to be used every time the students close up on the set in the VCSST and subsequently on board.

Limitations	No references or assistance will be provided during the PC.
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

PO 003 – Perform Passive Aural Listening (PAL)

1. **Performance**: Perform Passive Aural Listening (PAL) (C0195)

2. **Conditions**:

   a. Given:

      (1) Chronoscope;
      (2) Supervision;
      (3) Workstations; and
      (4) References.

   b. Denied: Nil.

   c. Environment:

      (1) At sea;
      (2) Night or day; and
      (3) Any weather conditions.

3. **Standard**: The Submarine Acoustic Operator shall perform Passive Aural Listening (PAL) (C0195) by:

   a. Analyzing aural and visual characteristics (T0008); and

   b. Classifying sound sources using Passive Aural Listening (PAL) (T0007).

4. **Training Limitations**: ADAC facilities and PAL instructors are needed IOT deliver this PO.

EO 003.01– Perform Passive Aural Listening

Analyze Aural and Visual Characteristics (T0008)

Lesson	Conditions	Sequence
003.01 A	Classroom/Classified work station	After PC 002
Time	**Classification**	**Instructor**
10 x 50 Minute Periods	Secret	PO2-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Describe the PAL analysis process; 2) Describe aural characteristics; and 3) Describe the PAL classification process.	Interactive Lecture

Scenario	Nil
Activity Steps	1) Instructor will present the 003.01A lecture on the PAL analysis process and aural characteristics. 2) During and on completion of the PAL lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) Classroom; 2) Chronoscope; 3) Calculator; 4) Classified workstation; and 5) PP&S.	1) Lesson Material; and 2) CFCD 113 (D) Vol 2 AIG.

Notes to Instructor
Make sure to reiterate the student the importance of PAL onboard submarines as it is used constantly.

EO 003.01 - Perform Passive Aural Listening

Analyze Aural and Visual Characteristics (T0008)

Lesson	Conditions	Sequence
003.01 B	Classroom/Classified work station	After EO 003.01 A
Time	**Classification**	**Instructor**
24 x 50 Minute Periods	Secret	PO2-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Perform Passive Aural Listening (T0007).	EDI

Scenario	Nil
Activity Steps	1) Instructor provides an area of operation to give the students an idea of what countries are to be expected in the lab. 2) Students are given eight PAL labs with ten audio files per lab. 3) Students listen to each audio file and determine blade and shaft configuration, RPM counts, aural characteristics, significant events, transients and classification. 4) Instructor reviews the answers as a group after each lab and replays audio files.

Resources	References
1) Classroom; 2) Chronoscope; 3) Calculator; 4) Classified workstation; and 5) PPS.	1) Lesson Material; and 2) CFCD 113 D Vol 2.

Notes to Instructor
Make sure to reiterate the student the importance of PAL onboard submarines as it is used constantly.

EC 003.01 – Perform Passive Aural Listening

Scope	Conditions	Sequence
EO 003.01	Classroom ADAC	After 003.01B
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret Can/US/GBR	PO2-PO1 SONAR Op (Subs)

Scenario	You are sailing in the position of a Sound Room Watch keeper. The watch leader directs you to classify multiple contacts using PAL.
Assessment Steps	1) The student is to demonstrate an ability to use the concepts taught in EO 003.01. 2) Student is given the audio contacts and a questionnaire. Student is expected to classify ten contacts using PAL Student has 40 minutes to write the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) EC 003.01 Questionnaire; 2) EC 003.01 Answer Guide; 3) PP&S; 4) Classroom with Computers; 5) Chronoscope; and 6) Pal audio contacts.	1) Lesson Material; and 2) CFCD 113 (D) Vol2 AIG.

Notes to Instructor
This material is to be used every time the students operate a passive sonar in the VCSTT, at sea and during the OJPR.

Limitations	No assistance allowed. This knowledge will be consolidated during the OJPR.
Impact of Failure	Remedial Training as required and Rewrite.

PC 003 – Perform Passive Aural Listening

Scope	Conditions	Sequence
PO 003	Classroom ADAC	After EC 003.01
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Secret(Can/GBR/USA)	PO2-PO1 SONAR Op (Subs)

Scenario	You are at sea, in international waters, there is warship, merchant, submarine and fishing traffic in the area. There is moderate amount of biologic concentration in your area of responsibility.
Assessment Steps	1) The student is to demonstrate an ability to classify ten sound sources using the skills taught in PO 003. The student is to use a Chronoscope and any reference material normally available at sea. 2) Student is expected to identify blade and shaft configuration, rpm count, aural characteristics, transients and contact classification. 3) The instructor will provide ten acoustic signatures to the students. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) Classroom; 2) CFCD 113 (E) AIG; 3) Chronoscope; and 4) Osprey.	1) Lesson Material; and 2) CFCD113 (E) AIG.

Notes to Instructor
This skill will be used and built upon when performing the duties and responsibilities of a sonar operator on submarines.

Limitations	No assistance will be provided during the PC.
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

PO 004 – Contribute to intelligence gathering Operations

1. **Performance**: Contribute to intelligence gathering Operations (B0030)

2. **Conditions**:

   a. Given:

      (1) State board;
      (2) Supervision;
      (3) Recording equipment;
      (4) Logs;
      (5) PP&S;
      (6) Sonars; and
      (7) References.

   b. Denied: Nil.

   c. Environment:

      (1) At sea and alongside;
      (2) Night or day; and
      (3) Any weather conditions.

3. **Standard**: The Submarine Acoustic Operator shall contribute to intelligence gathering Operations (B0030) by:

   a. Maintaining operational logs (B0055) to include:

      (1) Maintaining state boards (B0015);
      (2) Annotating recording media (C0075) to include:
         i. Recording acoustic intelligence data.

      (3) Correlating records (T0010).

   b. Disseminating intelligence information (B0040).

4. **Training Limitations**: Nil

EO 004.01– Maintain Operational Logs

Logs and Records

Lesson	Conditions	Sequence
004.01A	Classroom	After PC 003
Time	**Classification**	**Instructor**
4 x 50 Minute Periods	Confidential	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Explain purpose of Logs and Records; 2) Describe the duties and responsibilities of members of the records organization; 3) Explain types of Logs and Records; 4) Explain required Log contents; 5) Explain intelligence gathering process; and 6) Explain acoustic intelligence transportation processes.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 004.01A lecture on logs and records. 2) On completion of the logs and records lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 004.01-02A Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson Material; 2) NCPM 403; 3) CSRM; 4) AXP1(E); and 5) NDSOD.

Notes to Instructor
Make sure to reiterate to the students that as Logs and Records is Confidential, any material they take from lectures for any reason is to be treated with the same classification

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EO 004.01 - Maintain Operational Logs

Correlate Records (T0010) via SIR 1000W

Lesson	Conditions	Sequence
004.01B	Classroom	After 004.01-02 A
Time	**Classification**	**Instructor**
2 x 50 Minute Periods	Confidential	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Identify capabilities and limitations of SIR 1000W; 2) Describe annotation of recording media; 3) Describe recording procedure of acoustic intelligence data; and 4) Describe submarine specific SIR 1000W functions.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 004.01B lecture on SIR 1000W operations, capabilities and limitations. 2) On completion of the SIR1000W lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) VCSTT/TWS; and 2) Operational SIR 1000.	1) Lesson Material; 2) NCPM 403; 3) CSRM; and 4) Sony SIR 1000 instruction manual 2nd edition.

Notes to Instructor
The students will have a basic knowledge of this equipment already as it is used in the fleet as well. The main purpose of this lecture is to familiarize them with the settings and capabilities specific to submarine operations. Especially emphasize that tape speed and voltage have a significant impact on the quality of the audio recorded for post analysis.

EC 004.01 – Maintain Operational Logs

Scope	Conditions	Sequence
EO 004.01	Classroom	After 004.01B
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Confidential	MS-PO1, SONAR Op (Subs)

Scenario	You have returned from sea and are packaging up the sound room ACINT gathered during the trip. You are part of the Records Collection organization on board the submarine at sea and alongside and are required to have an understanding of the duties and responsibilities of the members of that organization.
Assessment Steps	1) The student is to demonstrate the knowledge of the various logs and records that he/she will be responsible to maintain as well as the process used to transport those records for post analysis by answering the knowledge based questions correctly. 2) Student is given the scenario and questionnaire. Student is expected to answer the knowledge based questions correctly. The student has 30 minutes to complete the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) EC 004.01- questionnaire 2) EC 004.01 Answer key 3) PP&S 4) Classroom	1) Lesson Material 2) NCPM 403 3) NAVORD 4) CSRM 5) AXP 1 (E)

Notes to Instructor
This material is to be used every time the students conduct training in the VCSTT and subsequently on board.

Limitations	No references and no assistance will be provided during the EC.
Impact of Failure	Remedial Training as required and Rewrite.

PC 004 – Contribute to Intelligence Gathering Operations

Scope	Conditions	Sequence
PO 004	Classroom	After EC 004.01
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Confidential	MS-PO1, SONAR Op (Subs)

Scenario	You have returned from sea and are packaging up the sound room Acint gathered during the trip. You are part of the Records Collection organization on board the submarine at sea and alongside and are required to have an understanding of the duties and responsibilities of the members of that organization.
Assessment Steps	1) The student is to demonstrate an understanding of the various logs and records that he/she will be responsible to maintain as well as the process used to transport those records for post analysis. 2) Student is given the scenario. The student is expected to demonstrate the knowledge required to maintain, collect and transport logs and records in accordance with the information taught in PO 004 and NDSODs. The student will cover the duties and responsibilities of the members of the records collection team, records forms, log format, log content, record classification, and method of transportation. The student has 40 minutes to complete the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) PC 004 Questionnaire 2) PC 004 Answer Guide 3) PP&S 4) Classroom	1) Lesson Material 2) NCPM 403 3) NDSOD 4) CSRM 5) AXP 1 (E)

Notes to Instructor
This material is to be used every time the students conduct training in the VCSTT and subsequently on board.

Limitations	No assistance will be provided during the PC.
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

PO 005 – Maintain Submarine's Acoustic Signature

1. **Performance**: Maintain Submarines Acoustic Signature (T0009)

2. **Conditions**:

   a. Given:

      (1) Submarine;
      (2) Sonars; and
      (3) References.

   b. Denied: Nil.

   c. Environment:

      (1) At sea and alongside;
      (2) Night or day; and
      (3) Any weather conditions.

3. **Standard**: The Submarine Acoustic Operator shall maintain submarines acoustic signature (T0009) to by:

   a. Conducting noise rounds (C0130) to include:

      (1) Performing noise rounds (C0135); and
      (2) Recording noise defects.

4. **Training Limitations**: Nil.

EO 005.01 – Maintain Submarine's Acoustic Signature

Signature Reduction Organization (SRO)

Lesson	Conditions	Sequence
005.01 A	Classroom	After PC 004
Time	**Classification**	**Instructor**
4 x 50 Minute Periods	Unclassified	MS-PO1 Sonar Op (Subs)

Learning Objectives	Method
1) Describe purpose of SRO; 2) Identify Signature Reduction Organization positions; 3) Describe the duties and responsibilities of the SRO; 4) Recognize noise defects; and 5) Describe noise reduction procedure.	Interactive Lecture

Scenario	Nil.
Activity Steps	1) Instructor will present the 005.01A lecture on the signature reduction organization composition and responsibilities. 2) Instructor will provide examples and promote a group discussion on noise husbandry and its importance. 3) On completion of the signature reduction organization lecture, instructor will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) EO 005.01A Presentation; 2) Confirmation Questions; and 3) Classroom.	1) Lesson material; 2) NAVORD 3395-1; 3) NAVORD 3000-7; 4) CFCD 109B; and 5) VCSSO's.

Notes to Instructor
Make sure to reiterate to the students that noise husbandry on submarines is everyone's responsibility, not just the SRO.

EC 005.01 – Maintain Submarine's Acoustic Signature

Scope	Conditions	Sequence
EO 005.01	Classroom	After completion of EO 005.01A
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Unclassified	MS-PO1 SONAR Op (Subs)

Scenario	You are performing noise husbandry on board a submarine and are responsible for the monitoring and reduction of radiated noise. You are part of the Signature Reduction Organization on board the submarine at sea and alongside and are required to carry out your duties as a member of that organization.
Assessment Steps	1) The student is to demonstrate the knowledge of the duties and responsibilities of Noise Detection and Reduction taught in EO 005.01 by answering the knowledge based questions correctly. 2) Student is given the scenario and questionnaire. Student is expected to answer the knowledge based questions correctly. The student has 30 minutes to complete the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) EC 005.01 Questionnaire; 2) EC 005.01 Answer key; 3) PP&S; and 4) Classroom.	1) Lesson Material; 2) NCPM 403; 3) NAVORD 3395-1; 4) NAVORD 3000-7; and 5) VCSSOs.

Notes to Instructor
This material is to be used on board HMC Submarines.

Limitations	No references and no assistance allowed. This knowledge will be consolidated during the OJPR.
Impact of Failure	Remedial Training as required and Rewrite.

PC 005 – Maintain Submarine's Acoustic Signature

Scope	Conditions	Sequence
PO 005	Classroom	After EC 005.01
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Unclassified	MS-PO1 SONAR Op (Subs)

Scenario	You are performing noise husbandry on board a submarine and are responsible for the monitoring and reduction of radiated noise. You are part of the Signature Reduction Organization on board the submarine at sea and alongside and are required to carry out your duties as a member of that organization.
Assessment Steps	1) The student is to demonstrate an understanding of when noise reduction procedure are to be implemented, when different routine are to be carried out, the duties and responsibilities of the noise detection team members, why quiet states change, and what the change in quiet state means for the submarine. 2) Student is given the scenario. The student is expected to demonstrate the knowledge required to perform noise husbandry in accordance with CFCD 109 (B) and the corresponding NAVORDs. The student will cover the duties and responsibilities of the members of the signature reduction organization, quiet states, noise shorts, noise husbandry routines, sources of radiated noise and noise monitoring methods. The student has 30 minutes to complete the test. 3) The instructor may provide clarification of any questions that are not fully understood. 4) Instructor will deliver a 10 minute debrief to provide correct answers by reviewing the test with the group, reinforcing important and misunderstood material.

Resources	References
1) PC 005 Questionnaire; 2) PC 005 Answer Guide; 3) PP&S; and 4) Classroom.	1) Lesson Material; 2) NCPM 403; 3) NAVORD 3395-1; 4) NAVORD 3000-7; and 5) VCSSOs.

Notes to Instructor
This material is to be used on board HMC Submarines.

Limitations	No assistance will be provided during the PC.
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

PO 006 – Conduct Vibration Analysis

1. **Performance**: Conduct vibration analysis

2. **Conditions**:

   a. Given:

      (1) Supervision;
      (2) References;
      (3) VA test equipment;
      (4) VA software; and
      (5) Baseline readings.

   b. Denied: Nil.

   c. Environment:

      (1) At sea or alongside;
      (2) Any weather conditions; and
      (3) Night or day.

3. **Standard**: The Submarine Acoustic Operator shall conduct a vibration analysis by operating vibration analysis equipment (C0210) IAW AKTO QSP MITE 109364 EO 011.03 standard statement.

4. **Training Limitations**: Vibrations analysis facilities need to be booked and Vibration analysis Instructor need to be CFTPO to deliver this section of the AILO qualification.

Task List

Task #	Task Statement	Trg / No Trg	PO
A0005	Perform communications circuit functional checks (comm checks)	T	001
A0010	Operate communications equipment	T	001
A0025	Encode/Decode tactical information (with Authenticate communication transmissions)	T	002
A0030	Authenticate communication transmissions	T	002
B0005	Perform operator checks on equipment	T	001
B0015	Maintain state boards	T	004
B0020	Enforce Emission control (EMCON) restrictions	T	001
B0025	Adhere to Emission control (EMCON) restrictions	T	001
B0030	Contribute to intelligence gathering Ops	T	004
B0040	Disseminate intelligence information	T	004
B0055	Maintain operational logs	T	004
C0025	Operate ancillary acoustic equipment	T	002
C0035	Launch underwater expendables	T	002
C0040	Analyzing XBT/XSV Forms	T	002
C0050	Prepare underwater expendables for launch	T	002
C0055	Verify underwater expendables inventories	T	002
C0065	Analyze passive acoustic data	T	001
C0070	Track acoustic contacts	T	001
C0075	Annotate recording media	T	004
C0080	Classify passive acoustic data	T	001
C0115	Calculate acoustic sensor range information	T	001
C0130	Conduct noise rounds	T	005
C0135	Perform noise rounds	T	005
C0195	Perform Passive Aural Listening (PAL)	T	003
C0210	Operate vibration analysis equipment	T	006
C0280	Operate main frame sonar	T	001
C0285	Operate classification console	NT	N/A
F0010	Record acoustic intelligence data	T	004
T0001	Setup passive equipment(s)	T	001

Task #	Task Statement	Trg / No Trg	PO
T0002	Operate passive equipment(s)	T	001
T0003	Setup main sonar in active mode	T	001
T0004	Operate active equipment	T	001
T0005	Setup ancillary acoustic/oceanographic equipment	T	001
T0006	Operate ancillary acoustic/oceanographic equipment	T	001
T0007	Classify sound sources using Passive Aural Listening (PAL)	T	003
T0008	Analyze aural and visual characteristics	T	003
T0009	Control ship's acoustic signature	T	005
T0010	Correlate records	T	004
T0011	Identify acoustic signatures	T	001
C0151	Determine Oceanographic Effects on Submarine Acoustic Operations	T	001
B0036	Present oceanographic brief	T	001
T0012	Operate underwater comms	T	002
T0013	Operate echo sounders	T	002
T0014	Operate velocimeter	T	002
T0015	Operate underwater expendables	T	002
T0016	Track Acoustic Intercept and Ranging (AI&R)	T	001
T0017	Track Broadband	T	001
T0018	Track Narrowband	T	001
T0019	Conduct self-Noise Monitoring	T	001
T0020	Record Acoustic Intelligence (ACINT) data	T	001
T0021	Operate sonar logger	T	001
T0022	Operate Digital Data Collection System (DDCS)	T	001
A0035	Extract information from COMPLANS, EMCON Plans and Frequency Plans	NT	N/A
A0020	Perform Search and Rescue (SAR)/Distress/SUBSMASH communications procedures	NT	N/A
B0010	Plot formations, screens and dispositions	NT	N/A
B0050	Operate CS	NT	N/A
B0085	Conduct escape/rescue training	NT	N/A

Task #	Task Statement	Trg / No Trg	PO
C0085	Localize passive acoustic contacts	NT	N/A
C0150	Determine Oceanographic Effects on Acoustic Operations	NT	N/A
C0220	Operate One Man Control (OMC)	NT	N/A
C0190	Operate Low Pressure (LP) Air System	NT	N/A
C0015	Operate active sonar equipment	NT	N/A
C0020	Operate passive acoustic equipment	NT	N/A
C0030	Operate oceanographic equipment	NT	N/A
B0035	Display intelligence via UWW brief	NT	N/A
C0105	Correlate active/passive/non-acoustic information	NT	N/A
C0170	Assist with loading/handling of torpedoes	NT	N/A
E0005	Compile ASW tactical picture	NT	N/A

Record of Proceedings

Dates: 09 July 2018 - TBC

Location: CFB Stadacona, S17, Chris Saunder's Conference Room

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Smith	G.S.R.	PO1	00324.01	NTDC(A)
Matheson	P.A.R.	PO2	00324.01	NFS(A)
Matte	K.M.L.	Lt(N)	00211	NTDC(A)

Working Group Objective:

1. The objective of this working group are:

    a. Design and develop a course that is as much performance-based as possible by choosing proper methodologies;
    b. Ensuring the instruction is as close as possible to the real job by incorporating current war scenario(s); and
    c. Removing the outdated systems.

Task – No Train Decisions:

Task #	Task Statement	Substantiation
A0035	Extract information from COMPLANS, EMCON Plans and Frequency Plans	This task is not performed by the acoustic operator on Subs but by the Watch Leader (Master Sailor).
A0020	Perform Search and Rescue (SAR)/Distress/SUBSMASH communications procedures	This task is trained on the Sub BSQ course.
B0010	Plot formations, screens and dispositions	This task is not conducted on Subs.
B0050	Operate CS	WG members assumed that the Combat system meant the torpedo system. This is trained on the Sub BSQ course.
B0085	Conduct escape/rescue training	This task is trained on the Sub BSQ course.
C0085	Localize passive acoustic contacts	This task is not performed by the acoustic operator on Subs; it is done at the PO1 level.
C0150	Determine Oceanographic Effects on Acoustic Operations	This task is trained on the Sonar Op RQ-S3 course.
C0220	Operate One Man Control (OMC)	This task is trained on the Sub BSQ course.
C0190	Operate Low Pressure (LP) Air System	This task is trained on the Sub BSQ course.
C0285	Operate classification console	This piece of equipment is outdated.
C0015	Operate active sonar equipment	Task cover by task T0004
C0020	Operate passive acoustic equipment	Task cover by task T0002
C0030	Operate oceanographic equipment	Task cover by task T0006
B0035	Display intelligence via UWW brief	This task is not perform at the S1 level

C0105	Correlate active/passive/non-acoustic information	This task is not performed by the acoustic operator on Subs but by the Watch Leader (Master Sailor).
C0170	Assist with loading/handling of torpedoes	Task covered in the weapon certification. Training is provided as needed by the boat to personnel that need the qualification: "just on time training".
E0005	Compile ASW tactical picture	Task cover by C0280 and this task statement is rather conducted at a higher level than the AILO qualification is training for.

Task added or modified:

Task #	Task Statement	Substantiation
C0151	Determine Oceanographic Effects on Submarine Acoustic Operations	This task is drawn from task C0150. Although the generic information for the Fleet Acoustic Operator have been covered on the Sonar Op QL3 course, there is specific information about how it applies to the Submarine that should be covered in the Sub QL4 course.
B0036	Present oceanographic brief	This task is drawn from task B0035. The Acoustic Op are delivery oceanographic brief.
C0280	Operate detection console / Operate main frame sonar	Task statement should be modify to Operate Main Frame Sonar to ensure members clearly understand what performance needs to be done.
T0012	Operate underwater comms	Gap were identified while scalaring. Acoustic Operators are required to operate underwater comms and no task was allowing us to cover this significant piece.
T0013	Operate echo sounders	Gap were identified while scalaring. Acoustic Operators are required to operate echo sounders and no task was allowing us to cover this significant piece.
T0014	Operate velocimeter	Gap were identified while scalaring. Acoustic Operators are required to operate velocimeter and no task was allowing us to cover this significant piece.
T0015	Operate underwater expendables	Gap were identified while scalaring. Acoustic Operators are required to operate underwater expendables and no task was allowing us to cover this significant piece.
B0005	Perform pre-sails checks on equipment / Perform operator checks on equipment	Task statement should be modify to perform operator checks on equipment to ensure members clearly understand what performance needs to be done and to encompass the entirety of the task.

T0016	Track Acoustic Intercept and Ranging (AI&R)	Gap were identified while scalaring. Acoustic Operators are required to Track Acoustic Intercept and Ranging (AI&R) and no task was allowing us to cover this significant piece.
T0017	Track Broadband	Gap were identified while scalaring. Acoustic Operators are required to Track Broadband and no task was allowing us to cover this significant piece.
T0018	Track Narrowband	Gap were identified while scalaring. Acoustic Operators are required to Track Narrowband and no task was allowing us to cover this significant piece.
T0019	Conduct self-Noise Monitoring	Gap were identified while scalaring. Acoustic Operators are required to Conduct self-Noise Monitoring and no task was allowing us to cover this significant piece.
T0020	Record Acoustic Intelligence (ACINT) data	Gap were identified while scalaring. Acoustic Operators are required to Record Acoustic Intelligence (ACINT) data and no task was allowing us to cover this significant piece.
T0021	Operate sonar logger	Gap were identified while scalaring. Acoustic Operators are required to Operate sonar logger and no task was allowing us to cover this significant piece.
T0022	Operate Digital Data Collection System (DDCS)	Gap were identified while scalaring. Acoustic Operators are required to Operate Digital Data Collection System (DDCS) and no task was allowing us to cover this significant piece.

Discussions / Decisions:

2. Sub Sonar Operator are require to complete vibration analysis at this level, as per task C0210, and are supposed to be loaded on the Vibrations analysis training (QUAL CODE AGSH) after successful completion of this course but they rarely are. It would be beneficial to the organization and submariner community, IOT resolve this situation, to merge the two qualifications together.

3. For safety reason, written ECs and the related written PC will be conducted prior to each trainer performance for PO 001. The written assessments enable verification of knowledge acquisition. Knowledge of the systems capabilities and limitations can the difference between life and death for the full crew on the boat. Therefore, boat safety will be maintain by making sure all capabilities and limitations of the main frame sonar are retained by each individual IOT ensure no fatal mistake can happen.

4. The BQQ-10 portion of the course is replacing the following systems: 2040, 2046, stand-alone PRS, 189, SIU, and soon, the SIR 1000W. Due to the size and intricacy of the system, as well as the integration of the vibration analysis training, the course duration will need to be extended.

Guidance to Development and/or Instructional Staff:

5. Nil.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	SOR submission	Sub CCD A/M	**14 Dec 2018**
2	Conduct Pilot Monitoring	CCD A / TDO / Instructor	**1 Nov 2018**

Course History:

6. **V2.0.01 dated 9 Aug 06**. The QSP was completely revised to incorporate the use of the VICTORIA-class NCOT as a training strategy.

7. **V2.01.02 dated 15 Feb 08**. After Session 0008, which was a pilot course, but due to manning issues it was not properly piloted. There were concerns raised by the instructor and comments from the student's critiques about PO 001 Operate Main Submarine Sonar Equipment. The major concerns were that there was too much material to understand (EO 001.2/EO 001.3) and there needed to be more practical time added to the course. After consultation with the instructor and the training development officer, it was recommended to add ten actual training days. The following changes were incorporated into Chapter 1 and Chapter 4:

a. Chapter 1, page1, paragraph 2, Course duration calculator actual training days changed from 23.2 days to 33.2. Training Administration Time was changed form 1.4 days to 1.9 days. Course duration total was changed from 26 to 36.5 days. Resources paragraph 3, page 5, critical training resources added Victoria Class Command Team Trainer.

b. Chapter 4, PO 001 Operate Main Submarine Sonar Equipment, EO 001.2 Operate the 2046 Flank/Towed Array Sonar, page 24, paragraph 4, Time- sub paragraph a, VCNCOT/CBT (TP1-29) was changed from 350 minutes to 1050 minutes. Time –sub paragraph e, Practical (TP31) was changed from 1050 minutes to 2100 minutes. Total was changed from 1850 min or 37 x 50 min period to 3600 min or 72 x 50 min periods

c. Chapter 4, PO 001 Operate Main Submarine Sonar Equipment, EO 001.3 Operate the 2040 Sonar and Intercept Classifier Unit, page 35, paragraph 4, Time- sub paragraph a, VNCOT/CBT (TP1-29) was changed from 350 minutes to 1050 minutes. Time- sub paragraph e, Practical (TP 27) was changed from 1050 minutes to 2100 minutes. Total was changed from 1850 min or 37 x 50 min period to 3600 min or 72 x 50 min periods

d. DMTE QSP Approval Submarine Sonar Operator Course 24 July 2008 was granted and is to have an effective date of 23 April 2008.RDIMS#142372

8. **V3.00 dated 18 Apr 11**. QSP approved by Capt. (N) Teft (D MAR PERS) 25 Mar 11.

a. DMTE 2-2-4 submitted the updated QSP, SPQR, and Change letter to D MIL Pers 25 Jan 2011, it was approved 24 Mar 11 (RDIMS# 205623). PO1 Martin (CFNOS STDS)

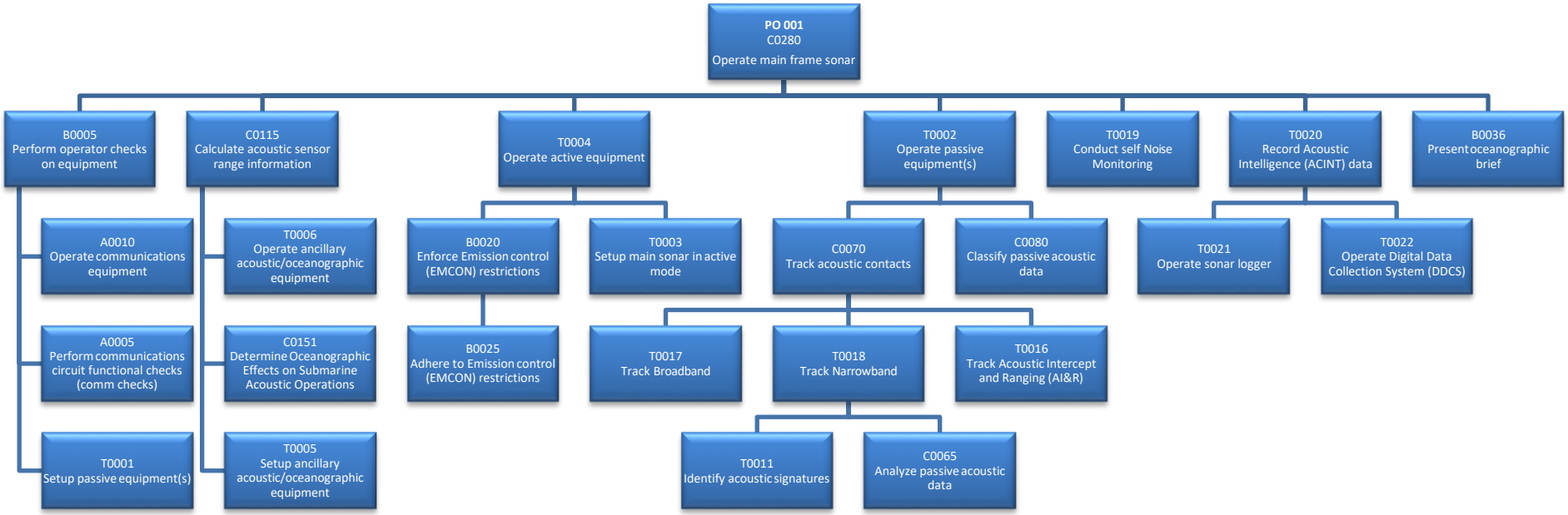
Incorporated the changes requested in the QSP CHANGE REQUESTS TO AILO SUBMARINE SONAR OP WATCHKEEPER, 12 Apr 11.

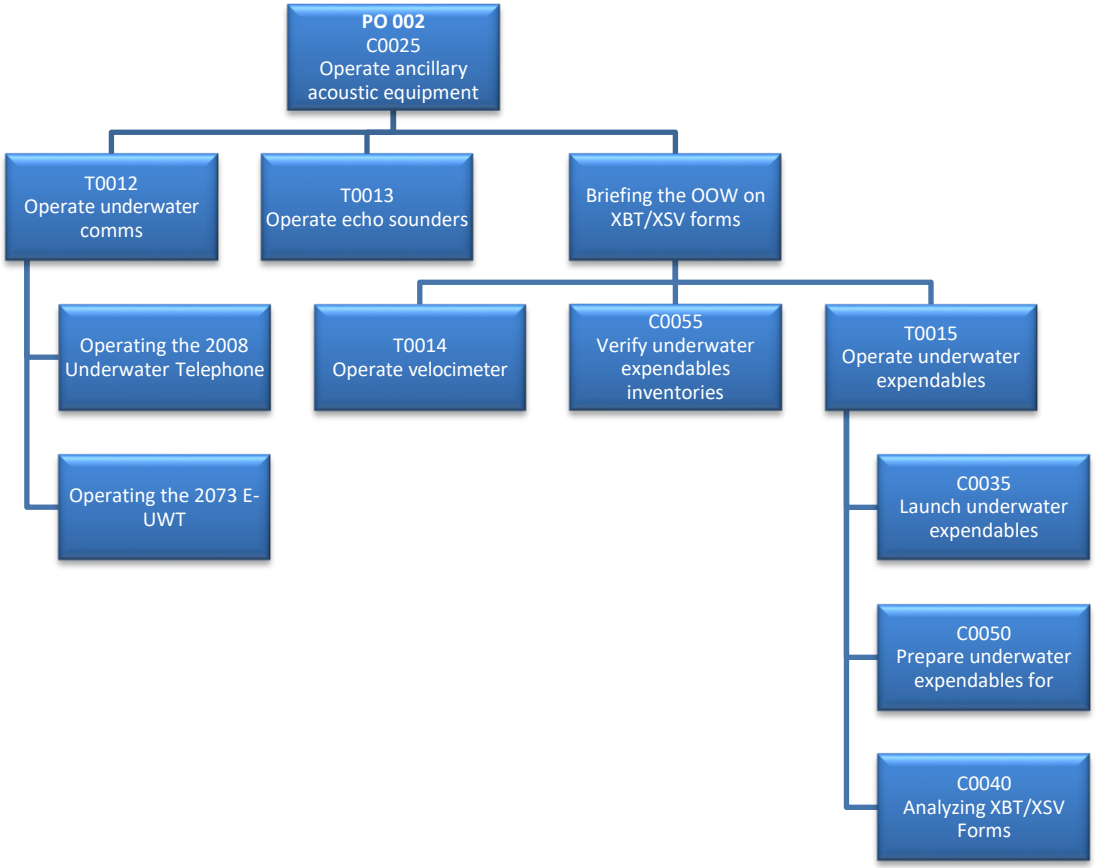
b. Updated QSP, with Sub-Version Date change submitted to Gerry Copithorn, DMTE 3-7-2-2 NTDSS Clerk for upload, 18 Apr 11 by PO2 Pierangeli, DMTE 2-2-4. Old version to be removed.

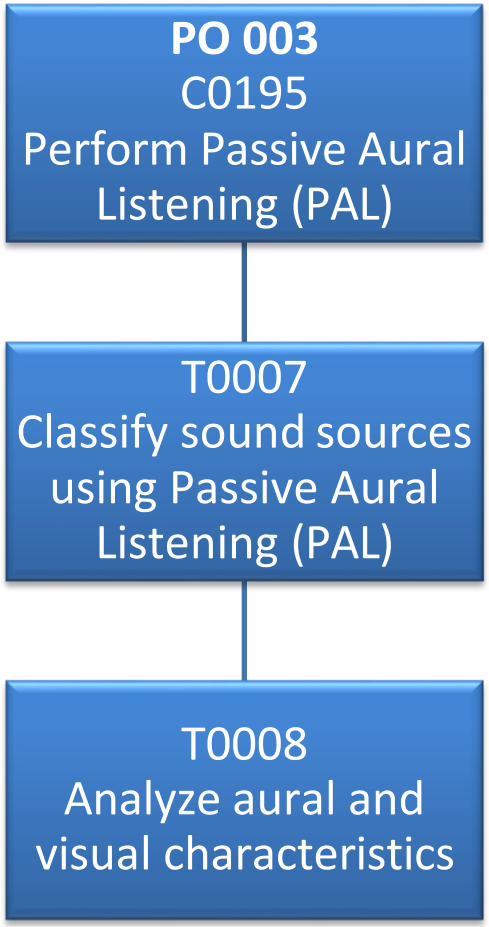
9. **V4.0 dated 30 Oct 15**. QSP has been reviewed by Sub Division Sonar Senior Instructor PO2 Lewis. There were many discrepancies found with the timings for various TP in the overall QSP. In order to make the QSP/course more efficient, timings have been reviewed and adjusted accordingly. The introduction of the BQQ-10 has also been inserted into the QSP, the reason for this is the VCSTT no longer supports the 2040 Sonar Suite and has been replaced by the BQQ-10 Sonar Suite. Sonar Op submariners will have to have an intimate knowledge of this new sonar suite before going on a Canadian Submarine and must be included in the train. The 2040 is still being used on 2 Canadian submarines, so this training has not been removed, it will be removed once the sonar suite is no longer on Canadian submarines. QSP was send for approval but due to a system glitch got lost.

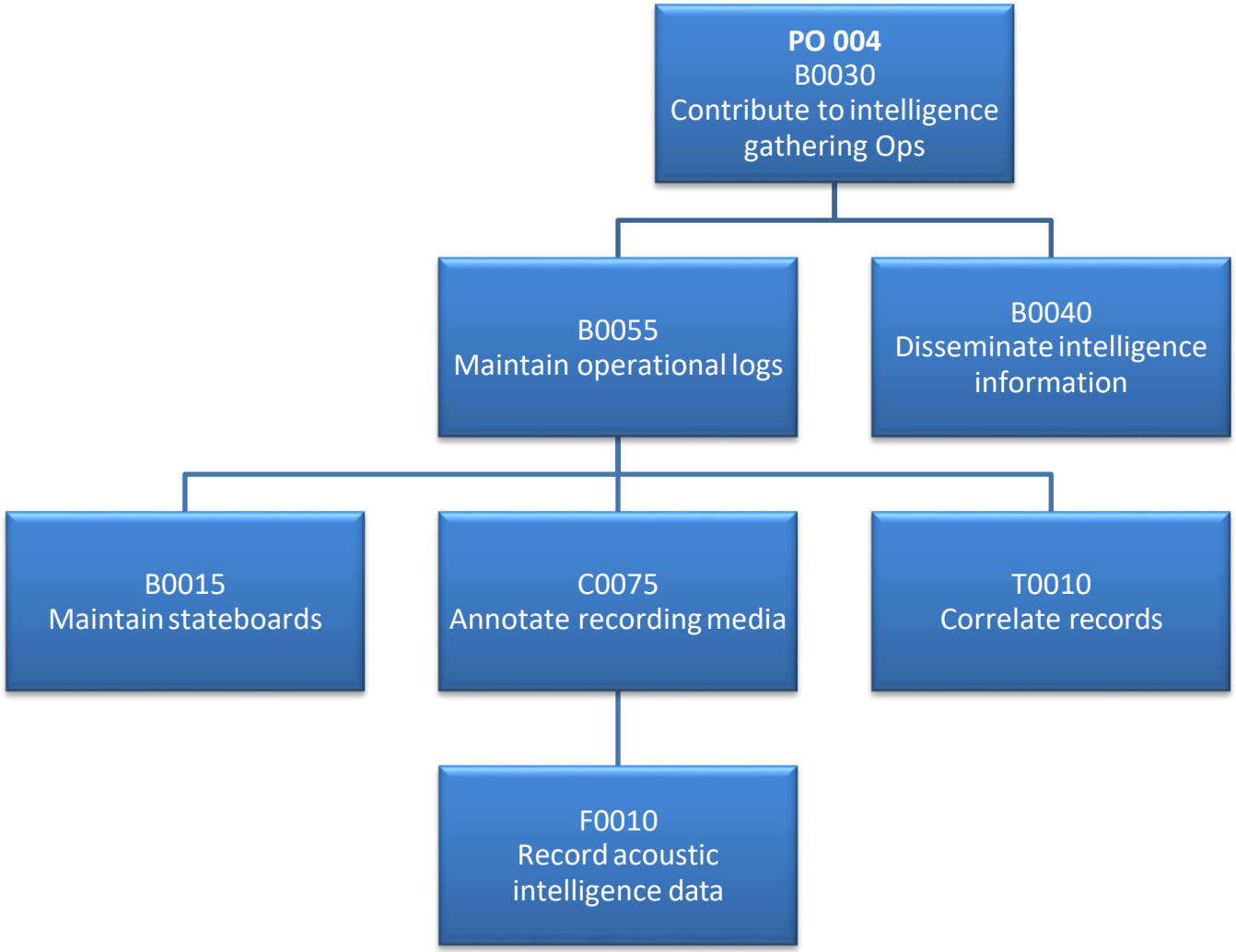
V5.1 dated 3 Feb 2021. A change in rank names was actioned IAW CANFORGEN 112/20 RCN Junior Rank Change Instructions.

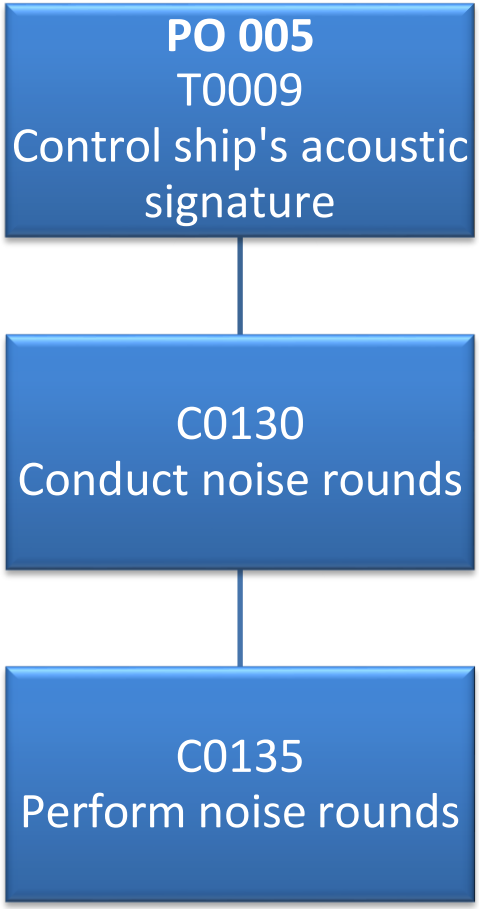
QSP Scalars

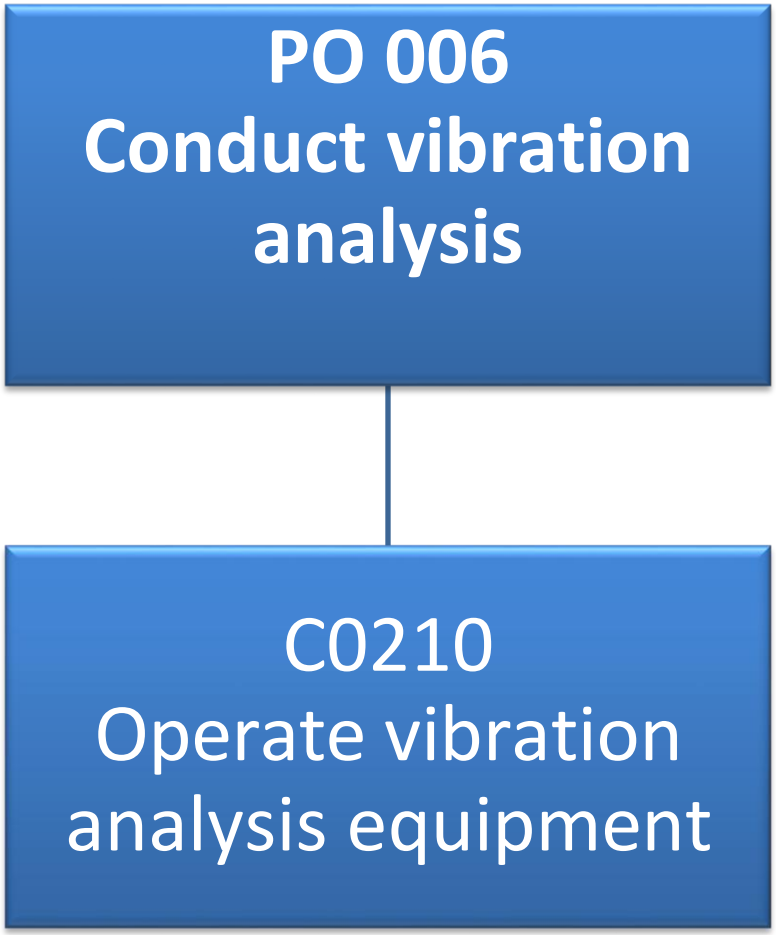

PO 001
C0280
Operate main frame sonar
B0005
Perform operator checks on equipment
C0115
Calculate acoustic sensor range information
T0004
Operate active equipment
T0002
Operate passive equipment(s)
T0019
Conduct self Noise Monitoring
T0020
Record Acoustic Intelligence (ACINT) data
B0036
Present oceanographic brief
A0010
Operate communications equipment
T0006
Operate ancillary acoustic/oceanographic equipment
B0020
Enforce Emission control (EMCON) restrictions
T0003
Setup main sonar in active mode
C0070
Track acoustic contacts
C0080
Classify passive acoustic data
T0021
Operate sonar logger
T0022
Operate Digital Data Collection System (DDCS)
A0005
Perform communications circuit functional checks (comm checks)
C0151
Determine Oceanographic Effects on Submarine Acoustic Operations
B0025
Adhere to Emission control (EMCON) restrictions
T0017
Track Broadband
T0018
Track Narrowband
T0016
Track Acoustic Intercept and Ranging (AI&R)
T0001
Setup passive equipment(s)
T0005
Setup ancillary acoustic/oceanographic equipment
T0011
Identify acoustic signatures
C0065
Analyze passive acoustic data


PO 002
C0025
Operate ancillary acoustic equipment
T0012
Operate underwater comms
T0013
Operate echo sounders
Briefing the OOW on XBT/XSV forms
Operating the 2008 Underwater Telephone
Operating the 2073 E-UWT
T0014
Operate velocimeter
C0055
Verify underwater expendables inventories
T0015
Operate underwater expendables
C0035
Launch underwater expendables
C0050
Prepare underwater expendables for
C0040
Analyzing XBT/XSV Forms


PO 003
C0195
Perform Passive Aural Listening (PAL)
T0007
Classify sound sources using Passive Aural Listening (PAL)
T0008
Analyze aural and visual characteristics


PO 004
B0030
Contribute to intelligence gathering Ops
B0055
Maintain operational logs
B0040
Disseminate intelligence information
B0015
Maintain stateboards
C0075
Annotate recording media
T0010
Correlate records
F0010
Record acoustic intelligence data


PO 005
T0009
Control ship's acoustic signature
C0130
Conduct noise rounds
C0135
Perform noise rounds


PO 006
Conduct vibration analysis
C0210
Operate vibration analysis equipment

Costing Information

Include information about cost in whichever format is applicable (**Not to be deleted**).

Costing Annex_Template.xlsx

Estimates only, based on CF Cost Factors Manual FY 2012/13 verify that this link is the most updated version

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: AILO	Serial:					
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	6	$1 000,00		$6 000	$6 000	$60 000,00
R&Q	6	$36,00	0	$0	$0	$0,00
Incidentals	6	$17,50	0	$0	$0	$0,00
Subtotal				**$6 000,00**	**$6 000,00**	**$60 000,00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	6	$50,00	1	$300	$300	$3 000,00
Classroom (square meters)	6	$13,00	47	…	…	…
Computer requirements	6	$1 000,00	1	$6 000	$0	$6 000,00
IT Equip (Proj, Smart board etc)	6	$6 000,00	1	$36 000	$0	$36 000,00
DWAN connection (Set up)	6	$5 000,00	1	$30 000	$0	$30 000,00
Link Toolkit (on DVD)	6	$1,00	1	$6	$15	$141,00
Instructional Materials (Printed / Handouts etc)	6	$30,00	1	$180	$450	$4 230,00
Subtotal				**$72 486**	**$765**	**$79 371,00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1) (incremental)5	1	$206,00	0	$0	$0	$0,00
Instructor (Lt(N)) (incremental)5	1	$235,00	0	$0	$0	$0,00
Incremental Civilian Instructor	0	0	0	$0	$0	$0,00
		subtotal		**$0,00**	**$0,00**	**$0,00**
	Total Estimated Costs		**$78 486**		**$6 765**	**$139 371**

Assessment Plan

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.01	To assess student's knowledge of operator checks.	Written	Trainee must achieve 70% IAW EC 001.01 solution.
EC 001.02	EO 001.02	To assess student's knowledge of capabilities and limitations of Sonar Tactical Decision Aids (STDA).	Written	Trainee must achieve 70% IAW EC 001.02 solution.
EC 001.03	EO 001.03	To assess student's knowledge of capabilities and limitations of active equipment.	Written	Trainee must achieve 70% IAW EC 001.03 solution.
EC 001.01-03	EO 001.01-03	To assess student's competence in performing operator checks on equipment	Practical	Trainee must achieve 30/39 SATs IAW EC 001.01-03 Checklist
EC 001.04	EO 001.04	To assess student's knowledge of capabilities and limitations of passive equipment.	Written	Trainee must achieve 70% IAW EC 001.04 Checklist
EC 001.05	EO 001.05	To assess student's competence in conducting self noise monitoring.	Written	Trainee must achieve 70% IAW EC 001.05 solution.
EC 001.06	EO 001.06	To assess student's competence in record ACINT data.	Written	Trainee must achieve 70% IAW EC 001.06 solution.
EC 001.04-06	EO 001.04-6	To assess student's competence operating passive, NM, and DDCS Sub-systems.	Practical	Trainee must achieve 84/112 SATs AND SATs on all critical items IAW EC 001.04-06 checklist.
EC 001.07	EO 001.07	To assess student's competence in performing oceanographic brief.	Practical	Trainee must achieve 12/16 SATs IAW EC 001.07 Checklist.
PC 001 A	PO 001	To assess student's knowledge of main frame sonar capabilities and limitations.	Written	Trainee must achieve 75% IAW PC 001A solution.
PC 001 B	PO 001	To assess student's competence in operating main frame sonar.	Practical	Trainee must achieve 114/151 SATs AND SATs on all critical items IAW PC 001B checklist.
EC 002.01	EO 002.01	To assess student's ability setting up the underwater telephones and operating them.	Practical	Trainee must achieve 13/18 SATs IAW EC 002.01 Checklist.
EC 002.02	EO 002.02	To assess student's knowledge of the capabilities and limitations of the 778/780 Echo Sounders.	Written	Trainee must achieve 70% IAW EC 002.02 solution.
EC 002.03	EO 002.03	To assess student's knowledge of the capabilities and limitations of the 2004 and MK8F.	Performance based written.	Trainee must achieve 75% IAW IAW EC 002.03 solution.

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
PC 002	PO 002	To assess student's knowledge of the capabilities and limitations of the ancillary sonars.	Written	Trainee must achieve 75% IAW PC 002 solution.
EC 003.01	EO 003.01	To assess student's competence in performing PAL	Performance based written	Trainee must achieve 75% IAW EC 003.01 solution.
PC 003	PO 003	To assess student's competence in performing PAL	Practical	Trainee must achieve 70% IAW PC 003 solution.
EC 004.01	EO 004.01	To assess student's competence in contributing to intelligence gathering Ops	Written	Trainee must achieve 70% IAW EC 004.01 solution.
PC 004	PO 004	To assess student's competence in contributing to intelligence gathering Ops	Written	Trainee must achieve 70% IAW PC 004 solution.
EC 005.01	EO 005.01	To assess student's knowledge of noise rounds	Written	Trainee must achieve 70% IAW EC 005.01 solution.
PC 005	PO 005	To assess student's knowledge of noise reduction procedures.	Written	Trainee must achieve 70% IAW PC 005 solution.
EC 006.01	EO 006.01	To assess student's competence in operating vibration analysis equipment	Practical	IAW AKTO QSP MITE 109364 EO 011.03
PC 006	PO 006	To assess student's competence in operating vibration analysis equipment	Practical	IAW AKTO QSP MITE 109364 EO 011.03

TEL Development Opportunities

Title	Type	Description of Requirement	Lesson	SOR #
Operating Procedures on all Sensors	Video (SECRET)	Classified video processing and creation facilities will greatly increase training effectiveness.	PO 001, PO 002, PO 004.	
778 trainer update	Trainer	Update 778	EO 002.02	
780 trainer update	Trainer	Update 780	EO 002.02	
2004 trainer update	Trainer	Update 2004	EO 002.03	
Virtual Submarine update	Virtual reality	Update virtual submarine to include noise monitoring functions	EO 005.01A	

Scheduling Information

Scheduling Annex (Firefly) v1.xlsx

Plan, Train, Do

PO	EO	Maintain Readiness	Plan	Train	Execute
001	01	Perform Operator Checks on Equipment			
	02	Calculate Acoustic Sensor Range Information			
	03			Operate Active Equipment	
	04			Operate Passive Equipment	
	05	Conduct Self Noise Monitoring			
	06			Record Acoustic Intelligence (ACINT) Data	
	07		Present Oceanographic Brief		
002	01			Operate Underwater Comms	
	02			Operate Echo Sounders	
	03		Analyze XBT/XSV Forms		
003	01			Perform Passive Aural Listening (PAL)	
004	01			Maintain Operational Logs	
	02				Disseminate Intelligence Information
005	01	Conduct Ship's Acoustic Signature			
006	01	Conduct Vibration Analysis			

References

A = CF or other official Canadian government publication
B = Official Allied Force or Allied government publication
C = Commercial Publication
D = Other

Ref Code	Publication
A1	NCPM 403
A2	AN/BQQ-10(V)7 TI10 APB09 Interactive Electronic Technical Manual
A3	C-69-827-000/MS-00 2046 Outboard Handbook and Parts List
A4	C-69-848-000/MS-001 2046 Inboard Handbook and Parts List
A5	C-69-846-000/MS-001 2008 Manual
A6	C-69-860-000/MS-001 2073 Handbook
A7	C-57-803-000/MS-001 778 Manual
A8	C-57-809-000/MS-001 780 Manual
A9	C-69-852-000/MS-001 2004 Handbook
A10	C-69-852-000/MA-Z01 2004 Handbook Supplement
A11	Canadian Submarine Records Manual
A12	National Defense Security Orders and Directives
A13	NAVORD 3395-1
A14	NAVORD 3000-7
A15	CFCD 109 (B)
A16	VCSSOs
B1	AXP 1 (E)
B2	BR 8070
C1	Lockheed Martin MK 8F information package
C2	Lockheed Martin AN/BQQ-10 (V)-7 TI-10 Sonar System Instructor Guide

On Job Performance Record (OJPR)

(CERTIFICATE OF QUALIFICATION)

_____

NAME and INITIALS RANK SN

_____

Date Training Started Expected Date of Completion

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, will be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. A duplicate copy will be made for retention by the trainee for the purpose of recording and reviewing individual progress.

2. As the trainee complete each Performance Objective (PO) to the required standard, the Training Petty Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. All attempts must be made to conduct OJT at sea. If this is not possible, these POs may be completed alongside and annotated in the remarks column. The starred designated items in the Performance Checklist must be completed at sea.

4. The completion of this OJPR requires 21 days at sea.

5. Upon completion of the OJPR sheet (trainee has completed all required POs), the following actions will be taken:

   a. The Divisional Officer and Commanding Officer shall complete section 8 of the OJPR Certificate of Qualification.

   b. A message will released from the unit stating the package has been complete. The message will be address to the following:

      i. NFS(A) Sub Training Div for entry into MITE (for Submarine Division Courses only);
      ii. Appropriate Career Manager;
      iii. DMTE4; and
      iv. N11-31 (MARLANTHQ and/or MARPACHQ).

PROGESS REVIEW

6. Progress review shall be carried out by the Divisional Officer and Section Head

OJPR REVIEWED BY (RANK, POSITION, NAME)	SIGNATURE	DATE

Add as many rows as necessary for this function.

TRAINING COMPLETION CERTIFICATE

7. Date training Completed: _____

_____

Section Head Date

CERTIFICATION OF QUALIFICATION

8. I certify that_____has completed Training to the standards stated in QSP-SONAROP SUBS and has achieved the AILO qualification.

_____

Commanding Officer Date

ON-JOB PERFORMANCE RECORD SHEET

1. The remarks column in this annex shall be used for the following information only:

   a. The unit which the training and assessment was carried out; and
   b. Any unusual circumstances which preclude compliance to an objective.

PO/EO	PERFORMANCE STATEMENT	DATE	SIGNATURE	REMARKS (INCLUDE UNIT)
001	Employ mainframe submarine sonar equipment.			
002	Employ ancillary Sonar equipment.			
003	Perform PAL	N/A	N/A	Standard met at NFS(A)
004	Maintain Logs and Records			
005	Maintain submarine's acoustic signature			
006	Perform Vibration Analysis	N/A	N/A	Standard met at NFS(A)

PERFORMANCE CHECKLIST

EO 001.01 AN/BQQ-10 Sonar System			
Performance Statement	**Date**	**Signature**	**Comments**
1) BQQ-10 closing up procedures, to include: a) SONSYM audit; b) Recorder status; c) Display settings; d) Sensor inputs; e) Update rates; f) Vertical beam width; and g) Audio settings.			
2) BQQ-10 standard operating procedures, to include: a) Report a new contact; b) Report a regained contact; c) Close Quarters Drill; d) Torpedo Drill; e) Emergency go deep procedure; f) Return to periscope depth; and g) Long post diving checks.			
3) BQQ-10 Components and Functions to include: a) Passive Arrays; b) Active Array; c) Processing Cabinets; d) 8037 Data Recorder; e) System power; f) System cooling; g) Daily user checks; h) Power supplies; i) Bearing coverage of arrays; j) Transmit frequencies; and k) Frequency band coverage.			

EO 001.02 Operate Sonar Tactical Decision Aids (STDA)			
Performance Statement	**Date**	**Signature**	**Comments**
1) STDA closing up procedures, to include: a) In-situ; b) Mission planning; and c) Select units of measurement.			*

2) STDA standard operating procedures, to include: a) Update SVP; b) Display SVP at multiple depths; and c) Manipulate map.			

EO 001.03 Operate HF Active Displays and Controls			
Performance Statement	**Date**	**Signature**	**Comments**
1) HFA closing up procedures, to include: a) Reconfiguration of sail string; b) Screen selection; c) Mode selection; d) Pulse selection; and e) Display configurations.			
2) HFA standard operations, to include: a) Transmit in IKA-Depth; b) Transmit in IKA-Fade; c) Transmit in MA; d) Transmit in ASW; e) Transmit in PD; f) Transmit in PUMA; g) Direct transmission via STDA thumbnail; h) Transmit in traditional and monopulse; and i) Conduct an Urgent Active Drill.			
3) BQQ-10 High Frequency Active operating capabilities to include: a) Sub-modes; b) HFA display set-up; c) SONSYM Configuration; d) HFA tracker assignment; and e) CO's transmit inhibit key.			

EO 001.04 Operate BQQ-10 Passive Sub-Systems			
Performance Statement	**Date**	**Signature**	**Comments**
1) CF promotion and functions, to include: a) Promote a CF; b) Manual tracking; c) CF editing; d) Feature CF interrogation; and e) Trace attributes.			
2) BQQ 10 Passive broadband operations : a) Carry out standard listening procedures; b) Identify a contact;			

c) Report a contact; d) Track a contact in auto; e) Track a contact in manual; f) Classify a contact; g) Conduct lost contact procedures; h) Identify contact Zig and i) Perform proximity checks on all sensors.			
3) Towed array operating procedures to include: a) Report a contact; b) Track a contact; c) Solve for ambiguity; d) Classify a contact; e) Conduct lost contact procedures; f) Regain a contact; and g) Conduct proximity checks.			
4) BQQ-10 passive narrowband operating procedures to include: a) Carry out a detailed narrowband search; b) Identify a contact; c) Correlate with broadband sensors; d) Promote a track to fire-control; e) Classify a contact; f) Identify indication of contact zig; g) Lost contact procedures; and h) Regain a contact.			
5) BQQ-10 Active Intercept and Ranging operating procedures to include: a) Identify transmissions; b) Report transmissions; c) Promote a contact to fire-control; d) Assess situation and analysis display area; e) Set threat bands; f) Identify threat band transmissions; g) Correlate active transmissions with passive sensors; and h) Carry out a Ping Stolen Range.			

EO 001.05 Operate BQQ-10 Noise Monitoring Sub System			
Performance Statement	**Date**	**Signature**	**Comments**
1) NM sub-system operations and functions to include: a) Select input sensors; b) Select bandwidth;			

c) Input gram center frequency; and d) Correlate lofar gram information with amplitude gram.			

EO 001.06 Operate BQQ-10 DDCS Sub System			
Performance Statement	**Date**	**Signature**	**Comments**
1) DDCS sub-system functions and operations to include: a) Load a LTO-4 tape; b) Select input sensors; c) Lease a tape for scribe recorder (HFA); d) LTO-4 tape duplication; e) Continuous and event recording; f) Create and recall a screen capture; g) Write a screen capture to disk; and h) Manually input data into the sonar logger.			

EO 001.07 Present an Oceanography Brief to Command			
Performance Statement	**Date**	**Signature**	**Comments**
1) Present an oceanography brief to command to include the following: a) Best listening depth; b) Best evasion depth vs SM/WS; c) Explain SVP features; d) Extrapolate sound propagation; and e) Content is accurate.			

EO 002.01 Operate Underwater Telephone/Scanner(2008)			
Performance Statement	**Date**	**Signature**	**Comments**
1) 2008 capabilities and limitations, to include: a) Power supplies; b) Transducer layout; c) Frequencies; d) Bearing coverage; e) Bearing accuracies; f) Amplifier cooling; g) Power outputs; h) Range accuracy; and i) Modes of operation.			
2) 2008 closing up procedures, to include: a) Command authority; b) Power up/down procedure;			

c) Pre/post diving checks; d) Daily user checks; e) Communications checks; f) Checking ships head; g) Speaker and headset volume; h) Gain control; and i) Cursor operation.			
3) 2008 operating procedures, to include: a) Standard listening procedure; b) Mark snap ranging; c) Proximity checks; d) New contact procedure; e) Lost contact procedure; f) Close and dangerous contact procedure; g) Stern arc clearance procedure; h) Preps for returning to PD; and i) Returning to PD.			
EO 002.01 Operate Emergency Underwater Telephone 2073			
Performance Statement	**Date**	**Signature**	**Comments**
1) Escape sonar components and functions, to include: a) Transducers; b) Sub to ship/sub communications; c) Sub to DSRV communications; d) Sub to diver communications; e) HF Ping; f) LF Ping; and g) Sonar Sound Telegraphy (SST).			
2) Escape sonar Capabilities and Limitations, to include: a) Power supplies; b) Transmit frequencies; and c) Range.			
3) Escape sonar operating procedures, to include: a) Diving checks; b) When to operate LF/HF Ping; c) Emergency operating procedure; and d) Operating equipment during: i) Long post diving checks; ii) Inshore ops; and iii) Attack/action stations.			*

EO 002.02 Operate Echo Sounders 778 / 780			
Performance Statement	**Date**	**Signature**	**Comments**
1) Echo sounder components and functions, to include: a) Transducers; b) Change over switches; and c) Equipment cabinets.			
2) 778/780 Capabilities and Limitations, to include: a) Power supplies; b) Frequencies; c) Depth scales; d) Power outputs; e) Functions and controls; and f) Modes of operation.			
3) Echo sounder closing up procedures, to include: a) Command authority; b) Navigation brief; c) Echo sounder matrix/policy; d) Power up/down procedure; e) Operator checks; f) Check paper; g) Echo sounder log; h) Charted depth; i) Stylus checks; j) Change over switches; and k) Digital readout.			
4) Echo sounder operating procedure, to include: a) Snap sounding; b) Normal operation; c) Paper change; d) No sounding procedure; e) Yellow and Red sounding reports; f) Changing stylus; and g) Record keeping.			*

EO 002.03 Operate Bathythermograph			
Performance Statement	**Date**	**Signature**	**Comments**
1) MK 8F Capabilities and Limitations, to include: a) SSXBT stowage; b) Launch requirements; c) Operating depth;			

d) Probe types; and e) Power supply.			
2) MK 8F Bathy operating procedures, to include: a) Set up the laptop to include updated position (LAT/LONG) and probe type; b) Load and fire an SSXBT/SV; and c) Recognition of U/S bathy trace.			*
3) 2004 Velocimeter components and functions, to include: a) Sound head locations; b) Depth scale; c) Depth bracket; d) Recorder; and e) Depth sensor vent.			
4) 2004 Velocimeter operating procedures, to include: a) Selection of sound head; b) Instillation of card; c) Calibration; d) Venting procedure; e) Reporting procedure; f) Major depth changes; and g) Returning to PD.			*

EO 004.01 Maintain Logs and Records			
Performance Statement	**Date**	**Signature**	**Comments**
1) Maintain logs and records to include: a) Bathy Log; b) Echo Sounder Log; c) UWT Log; d) Sound Room Narrative(When Required); e) Augmenter Log; f) Noise Rounds Log; g) Noise Defect Log; and h) Artefact Log.			
2) Package logs for transport to post analysis facility IAW NDSODs.			

EO 005.01 Signature Reduction Organization			
Performance Statement	**Date**	**Signature**	**Comments**
1) Conduct noise rounds.			
2) Recognize noise shorts.			
3) Duties and responsibilities of the SRO to include: a) Signature Reduction Officer; b) Noise Detection Officer; c) Noise Engineering Officer; d) Fleet Technical Authority; e) Casing Officer; and f) Noise Detection Team.			

*** NOTE: The starred items must be completed at sea.**

Complete and Pass 5 Monthly Training Packs			
Performance Statement	**Date**	**Signature**	**Comments**